

Regression Discontinuity Design with Multivalued Treatments *

Carolina Caetano Gregorio Caetano
University of Georgia *University of Georgia*

Juan Carlos Escanciano[†]
Universidad Carlos III de Madrid

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Abstract

We study identification and estimation in the Regression Discontinuity Design (RDD) with a multivalued treatment variable. Assuming linearity in this context can be unrealistic. For example, in the estimation of the effect of Medicare insurance coverage on health care utilization, the marginal effect of having Medicare as a first insurance is likely very different from the marginal effect of having it as a second insurance. We show that heterogeneity in the first stage discontinuities can be used for identification of nonlinear treatment effects under an alternative assumption, namely the homogeneity of the LATEs along some covariates. This assumption can often be tested and relaxed, and indeed we show that it seems to be valid in our context. We find that Medicare as a first insurance affects recurrent, lower cost health care utilization (e.g., doctor visits), while Medicare as a second insurance affects sporadic, higher cost health care utilization (e.g., hospital visits). Our estimator can be programmed as a simple Two-Stage Least Squares regression, and packaged standard errors and tests can also be used.

Keywords: Regression Discontinuity Design; Multiple Treatments; Over-identifying restrictions; Nonparametric; LATE; Medicare; Health Insurance.

JEL classification: C13; C14; C21; I13

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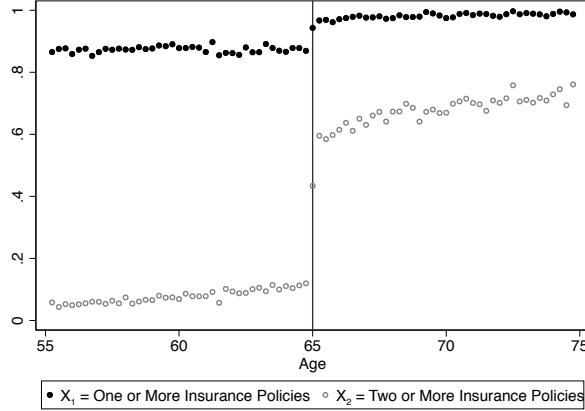
1 Introduction

Regression Discontinuity Design (RDD) has emerged as one of the most credible identification strategies in the social sciences. Most of the theoretical work on RDD focuses on the case where the treatment variable is binary. In this paper we study RDD for a non-binary treatment variable.

Applications where an RDD approach is available and the treatment variable is multivalued are quite prevalent in the social sciences. See, for example, [Angrist and Lavy \(1999\)](#), [Chay and Greenstone \(2005\)](#), [Ludwig and Miller \(2007\)](#), [Carpenter and Dobkin \(2009\)](#), [McCrary and Royer \(2011\)](#), [Brollo et al. \(2013\)](#), [Pop-Eleches and Urquiola \(2013\)](#), [Buser \(2015\)](#), [Isen et al. \(2017\)](#), [Spenkuch and Toniatti \(2018\)](#), [Nakamura and Steinsson \(2018\)](#), [Campante and Yanagizawa-Drott \(2018\)](#), [Corbi et al. \(2019\)](#), [Agarwal et al. \(2018\)](#), [Dell and Querubin \(2018\)](#), [Finkelstein et al. \(2019\)](#), [Dube et al. \(2019\)](#), and [Fort et al. \(2020\)](#), among others. In these settings, the standard RDD estimator identifies a weighted average of all marginal treatment effects among compliers for that treatment level, with weights proportional to the first stages of each treatment level ([Angrist and Imbens 1995](#)). Whenever there is the need to identify specific marginal effects, a linearity assumption is invoked (which is equivalent to assuming that all marginal effects are the same). However, linearity may be implausible.

To motivate our work, consider our application, where we want to identify the effect of different amounts of insurance coverage on healthcare utilization, as in [Card et al. \(2008\)](#). Medicare eligibility increases discontinuously at age 65, yet unobservables plausibly do not change discontinuously at age 65, which provides a useful quasi-experimental scenario. The issue is that Medicare eligibility should affect the outcome in very different ways depending on the margin: while for some people Medicare would be their first insurance, for others Medicare would be their second, more generous insurance. This can be seen in [Figure 1](#), which shows that both X_1 (an indicator of whether the person has one or more insurances) and X_2 (an indicator of whether the person has two or more insurances) are discontinuous at the 65 years of age

Figure 1: **First Stage**



Note: The plot shows, for each age (measured in quarters of a year), the percentage of people that have one or more insurances (black dots) and two or more insurances (hollow dots). See Section 5 for details.

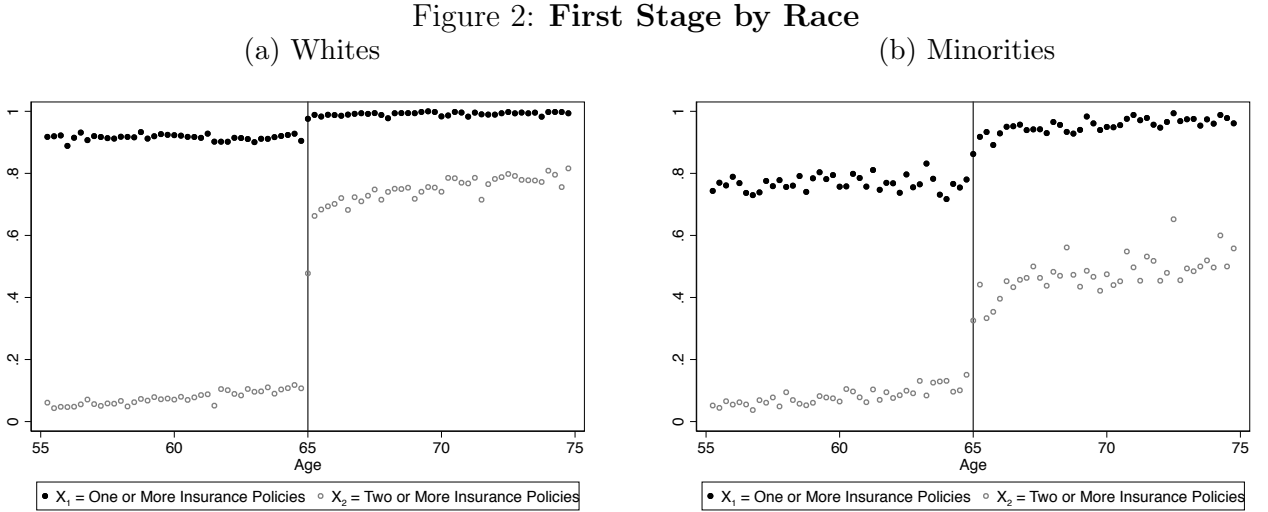
threshold.

In this setting, the standard RDD estimator with the number of policies as the treatment variable identifies the weighted average effect of Medicare coverage, which conflates the effect at the extensive margin (from no insurance to one insurance, Medicare) and the effect at the intensive margin (from one insurance, a private option, to two insurances, Medicare and a private option). If one wants to separately identify the effects at these two margins, there is currently no alternative inside the RDD framework. In particular, *ad hoc* strategies, such as implementing a standard RDD using only X_1 as the treatment, or restricting the sample only to those with zero or one insurance, do not achieve meaningful estimates, as we detail in Section 5.

In this paper, we propose a set of assumptions that enable the identification of different marginal effects in these multivalued treatment RDD settings. The resulting estimator is extremely simple, and implementation requires only a one line code for a 2SLS regression in packaged software. The packaged standard errors obtained from this regression can be used for inference under undersmoothing.

Figure 2 sheds light on the intuition of our method. The figure shows the discontinuities in both “having at least one insurance” (X_1) and “having at least two insurances” (X_2) separately for whites and minorities. We see that both variables are discontinuous in every instance.

We can write two reduced form equations, one for whites, and one for minorities, relating the discontinuities in the outcome to the first stage discontinuities in both treatment variables X_1 and X_2 . Thus, we have two equations with two unknowns: the marginal effect of X_1 , and the marginal effect of X_2 . Figure 2 also shows that *the first stage discontinuities across race are linearly independent*. Indeed, note that the ratio of the discontinuities in Figure 2 is different across panels: for whites, the discontinuity is much larger for the treatment variable X_2 while for minorities the discontinuities for the two treatment variables are of similar magnitude. This implies that the equations are linearly independent, and we can solve the system to identify the two unknown effects.



Note: The plot shows, for each age (measured in quarters of a year), the percentage of people with W as described in the plot caption that have one or more insurances (black dots) and two or more insurances (hollow dots). See Section 5 for details.

This example also clarifies the assumptions of this method. First, the discontinuities in the first stages per race must be linearly independent. This is easily verifiable as discussed above. Second, we use the separation into whites and minorities to generate two equations with two unknown parameters: the marginal effect of X_1 , and the marginal effect of X_2 . However, for there to be only two unknown parameters in this system, each of these two marginal effects must be the same for whites and minorities. This is what we call the homogeneity condition. Otherwise, the separation by race would yield two equations and four unknowns: two marginal

treatment effects per race.

We show how the homogeneity condition can be justified by the researcher, and even tested or relaxed. Moreover, we also study what our method identifies when this condition fails: a weighted average of all effects which minimizes the distance to the actual treatment effects, and has weights which add up to one on the effects of interest, and zero on the remaining effects. Thus, for example, the extensive margin coefficient of “having at least one insurance” minimizes the distance to the extensive margin effects for whites and minorities, and is a weighted average with weights that add up to one on the extensive margin of having insurance at all across the different races, and weights that add up to zero on the intensive margin of having generous insurance across the different races. This theoretical result is useful because it guarantees that our estimator is robust to small violations of the homogeneity assumption.

Using the National Health Interview Survey (NHIS) from 1999 to 2003, we find that the extensive margin of having insurance at all is the main reason for delaying or rationing care for cost reasons, resulting in a reduction of about 15 percentage points. The intensive margin of having more generous insurance is the main determinant of hospital visits, resulting on an increase of about 4 percentage points. Our results confirm the intuition suggested by [Card et al. \(2008\)](#) that having insurance at all affects recurrent, lower cost health care utilization (e.g., doctor visits), while having more generous insurance affects sporadic, higher cost health care utilization (e.g., hospital visits).

This paper adds to the growing literature on regression discontinuity designs. See [Imbens and Lemieux \(2008\)](#) and [Lee and Lemieux \(2010\)](#) for early surveys of the literature and [Cattaneo and Escanciano \(2017\)](#) and [Cattaneo et al. \(2019\)](#) for more recent overviews. As far as we know, [Dong et al. \(2021\)](#) is the only other paper that studies identification of RDD in non-binary treatment variables. The two papers are complementary, because they make different assumptions, and identify different quantities. [Dong et al. \(2021\)](#) relies on a rank-invariance assumption to identify quantile treatment effects when the treatment is continuous, while our approach relies on an homogeneity in covariates assumption to identify marginal treatment

effects when treatment takes multiple values. Our approach can also be used to separately identify the effects of multiple treatment variables in a linear model in the standard RDD setting (i.e. with a single running variable and threshold), see Remark 2.1.

Since identification in the RDD is conceptually similar to identification with a binary instrumental variable (IV), our paper is also related to that literature. Newey and Powell (2003) showed that rank conditions require the IV to be at least as complex as the number of parameters one wants to identify. This means that a binary IV can identify only a single slope parameter. Angrist and Imbens (1995) studied the interpretation of that parameter when the treatment is multivalued. With further assumptions, other papers proposed strategies for identification of marginal treatment effects with a binary IV (e.g., D’Haultfœuille and Février 2015, Torgovitsky 2015, Masten and Torgovitsky 2016, Huang et al. 2019 and Caetano and Escanciano 2021). We explore a similar condition to the covariance completeness assumption introduced in Caetano and Escanciano (2021), but here we offer a minimum-distance interpretation with the strongest possible identification that is particularly relevant in the RDD setting.

The remainder of the paper is organized as follows: Section 2 discusses identification, and Section 3 discusses the homogeneity assumption: how it may be justified, tested and relaxed, and how the approach behaves when it fails. Section 4 discusses estimation. Section 5 presents the application of our method to the problem of identifying the effects of different types of insurance coverage on health care utilization. Finally, we conclude in Section 6. An online appendix contains technical details about the justification of the homogeneity assumption, as well as a formal treatment of identification in the RDD with multivalued treatments and covariates, with proofs of the identification and asymptotic results.

2 Identification of Multiple Treatment Effects

In this section we present a minimalist exposition of the main arguments and results, assuming that the reader is familiar with the Regression Discontinuity Design (RDD) terminology and

ideas. We leave the rigorous treatment of the topic and all the details to Appendix B.

Suppose that there are $d + 1$ possible treatments, denoted t_j , where $t_0 < t_1 < \dots < t_d$. Let T_i denote the treatment level of individual i , and $Y_i(j)$ denote the potential outcome of person i to treatment j , so that the marginal effect of switching from one treatment to the next is $\beta_{ij} = Y_i(j) - Y_i(j - 1)$, and $\beta_i = (\beta_{i1}, \dots, \beta_{id})'$. Let $1(\cdot)$ denote the indicator function, and $X_i = (1(T_i \geq t_1), \dots, 1(T_i \geq t_d))'$, then the observed outcome is

$$Y_i = Y_i(0) + \beta_i' X_i. \quad (1)$$

We focus on this nonparametric setup with a finite number of treatment levels. Nevertheless, all our results also hold for model (1) with any finite vector X_i , e.g. $X_i = (Q_{1i}, Q_{2i}, Q_{3i})$, where Q_{1i} is the person's income, Q_{2i} is the daily number of cigarettes the person smokes, and Q_{3i} is an indicator of gender. See Remark 2.1.

Self-selection into treatment makes the vector of treatment indicators X_i potentially correlated with $Y_i(0)$ and β_i . We suppose that there exists a “running” variable Z_i and a threshold z_0 where an RDD approach may be possible conditional on a vector of covariates W_i . That is, we believe that the distributions of the unobservables are continuous across that threshold, while the distribution of X_i is discontinuous across the threshold, conditional on W_i .¹

Define $X_{ij}(z)$ the potential treatment level indicator function. That is, $X_{ij}(z) = 1$ if, whenever $Z_i = z$, $X_{ij} = 1$ (i.e. $T_i \geq t_j$), and analogously for $X_{ij}(z) = 0$. Define

$$\Delta_{ij} = \lim_{e \downarrow 0} (X_{ij}(z_0 + e) - X_{ij}(z_0 - e)).$$

A “complier” with respect to treatment j satisfies $\Delta_{ij} = 1$. To clarify, a complier for treatment level j is an observation i such that if Z_i is just below the threshold z_0 , $T_i < t_j$, and if Z_i is just above the threshold z_0 , $T_i \geq t_j$. Thus, this group includes all observations who would react

¹Precisely, see Assumption 3 in Appendix B.1, which is identical to the standard RDD assumptions, but for multiple treatments and conditional on covariates.

to the threshold incentive by increasing their treatment from any level below t_j to any level t_j and above. An observation that, upon crossing the running variable threshold, increases their treatment level more than one level is therefore a complier for more than one treatment level. For example, someone who would go from treatment t_1 to treatment t_3 is a complier for levels $j = 2$ and $j = 3$.

In our application, T_i represents the number of health insurance policies person i holds, X_{i1} is an indicator that i holds at least one insurance policy, and X_{i2} is an indicator that i holds at least two insurance policies. The running variable Z_i represents i 's age (measured in quarter increments), and the threshold $z_0 = 65$ represents the minimum age eligibility to Medicare. A complier for treatment level $j = 1$ is someone who would have no insurance if they were just a bit younger than 65 years of age, but would have one or more insurances if they were just a bit older than 65 years of age. Analogously, a complier for treatment level $j = 2$ is someone who would have one or no insurance if they were just a bit younger than 65 years of age, but would have two or more insurances if they were just a bit older than 65 years of age. Our data also has information on the person's race (white or minority), education (high school dropout or less, high school graduate, and some college) as well as information on gender, Hispanic ethnicity, and indicators of the region and year of the sample, all of which may be part of the control vector W_i . For simplicity we explore only $W_i = \text{race and/or education}$ in the illustrations, but we also use all covariates in the application.

Define the LATE $\beta_j(w) = E[\beta_{ij}|W_i = w, \Delta_{ij} = 1]$ as the average marginal effect of switching from treatment $j - 1$ to treatment j conditional on $W_i = w$, among compliers for treatment level j , at the threshold z_0 . In our application $\beta_1(w)$ represents the expected effect of going from no insurance to one insurance for those who are 65 years old, have $W_i = w$, and are compliers for treatment level 1. $\beta_2(w)$ is analogously defined.

For a placeholder variable V , denote $\delta_V(w) = \lim_{z \downarrow z_0} E[V|Z = z, W = w] - \lim_{z \uparrow z_0} E[V|Z = z, W = w]$, the discontinuity of V at the threshold conditional on $W = w$, and let $\beta(w)$ denote

the vector whose j th element is $\beta_j(w)$. Then, we obtain our main identification equation

$$\delta_Y(w) = \beta(w)' \delta_X(w), \quad (2)$$

for all w in the support of the distribution of W . Equation (2) follows from equation (1) and Assumption 3 in Appendix B, which makes requirements of continuity and monotonicity that are analogous to the standard RDD adapted to the conditional case.² The system of equations (2) relates first stage discontinuities $\delta_X(w)$ and the reduced form discontinuities $\delta_Y(w)$ with the structural parameters of interest $\beta_j(w)$, for each value w . Here, we are departing from the standard RDD approach by creating a system of equations, one equation for each value of w in the support of W 's distribution.

Let q denote the number of values that W assumes. The system of equations (2) has q equations and $q \cdot d$ parameters of interest. Therefore, the $\beta(w)$ are generally not identifiable. In fact, we show in Appendix B.1 that without further assumptions it is not even possible to identify interesting averages of the $\beta_j(W)$ separately from the $\beta_s(W)$ for $s \neq j$. In particular, it is not possible to identify the unconditional LATEs.³

Therefore, restricting the variation of the $\beta_j(w)$ is necessary for identification. Intuitively, we must have as many linearly independent equations as unknowns in the system of equations (2). Applications with multivalued treatment often impose restrictions across different values of j , that is, they explicitly or implicitly assume linearity (i.e., $\beta_j(W) = \beta_{j'}(W) \forall j, j'$ ⁴). When linearity is not plausible, we propose that identification be achieved through restrictions across different values of w instead. The strictest such restriction is to suppose the following homogeneity condition.

Assumption 1 $\beta_j(w) = \beta_j$, for all w and $j = 1, \dots, d$.

If $W = \text{race}$, this supposes that whites and minorities have the same LATEs, β_1 and β_2 . That is,

²The derivation of equation (2) can be found inside the proof of Proposition B.1 in Appendix C.

³See Appendix B.1 for a formal treatment of the identification problems in the multivalued RDD setting.

⁴We denote this assumption as linearity because when $t_j = j$ the main identification equation becomes a linear equation in T_i , i.e. $\sum_{j=1}^d X_{ij} = T_i$.

β_1 and β_2 are allowed to be different from each other in an unrestricted way, but $\beta_1(White) = \beta_1(Minority) = \beta_1$ and $\beta_2(White) = \beta_2(Minority) = \beta_2$. We discuss Assumption 1 in Section 3.

Given Assumption 1, we now have q equations and d LATEs we would like to identify. The ability to identify β depends on the rank condition:

Assumption 2 $E[\delta_X(W)\delta_X(W)']$ is positive definite.

Assumption 2, can be tested directly. Moreover, if W assumes a finite number of values, it is equivalent to a requirement that at least d among the first stage vectors are linearly independent.⁵ In our application, this means that all that is required for Assumption 2 to hold is that two of the conditional first stage vectors are linearly independent. This is obviously true for $W = \text{Race}$, as can be seen in Figure 2, in which the sizes of the discontinuities of X_1 and X_2 across the threshold for whites and minorities are clearly not proportional. For $W = (\text{Race}, \text{Education})$, it is sufficient to compare the first stage discontinuities for whites with some college (Figure 3(a)), with the first stage discontinuities for the minorities who are high school dropouts (Figure 3(f)), which are clearly not proportional.

If Assumptions 1 and 2 hold, equation (2) yields

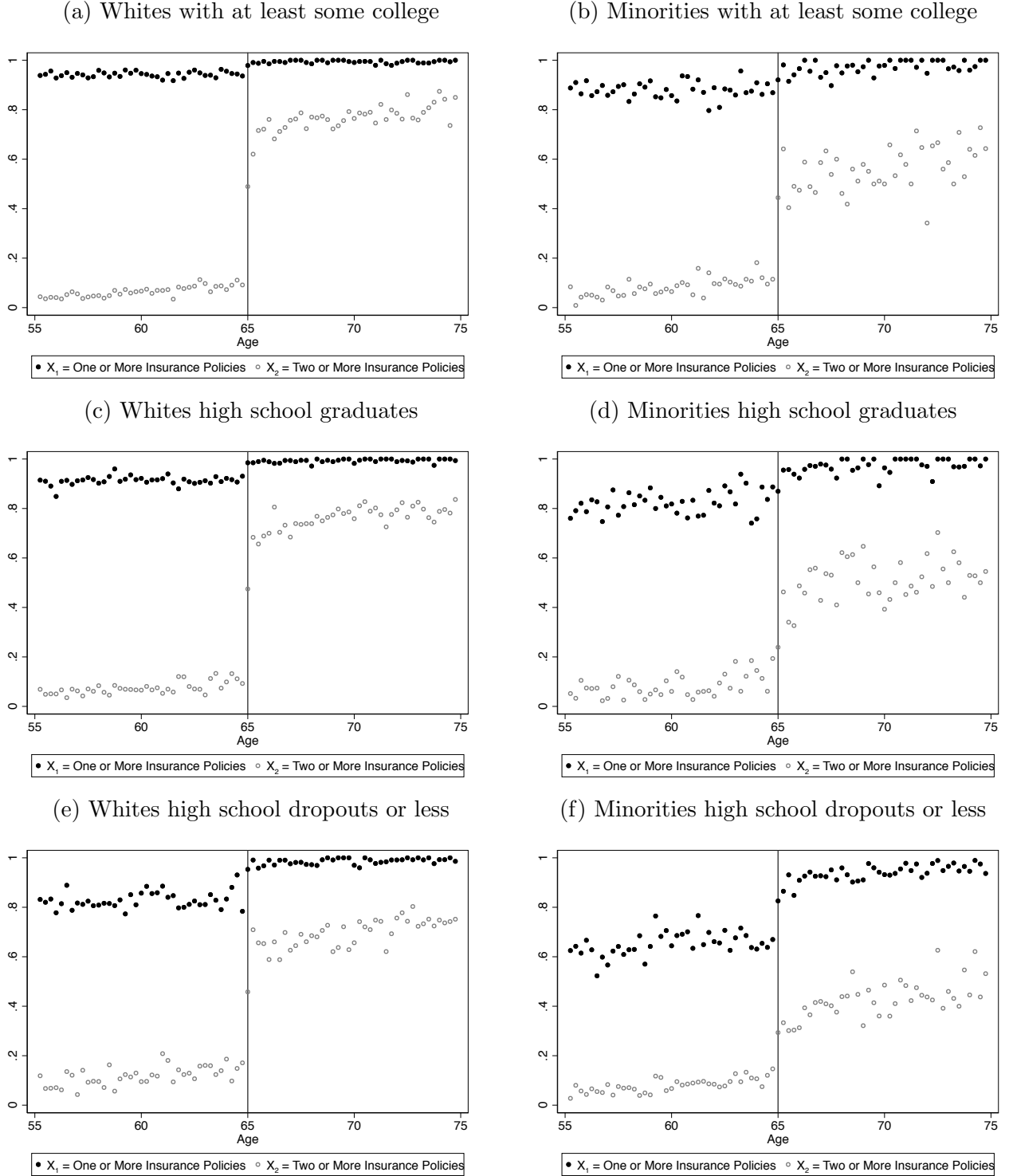
$$\beta = E[\delta_X(W)\delta_X(W)']^{-1}E[\delta_X(W)\delta_Y(W)']. \quad (3)$$

If W assumes d values (i.e., $q = d$), then β is just-identified. If W assumes more than d values ($q > d$), then β is over-identified. This allows us to test Assumption 1 (Section 3.2), and/or relax it, by assuming instead that $\beta(W)$ is linear in W , or another form with no more than q parameters (Section 3.3).

Estimation can be done by substituting the quantities above by sample equivalents. In Section 4 we show that, in fact, this can be implemented even more simply as a single Two Stage Least Squares regression.

⁵The proof of this statement is provided in Footnote 22 in Appendix B.2.

Figure 3: First Stage by Race and Education



Note: The plot shows, for each age (measured in quarters of a year), the percentage of people with W as described in the plot caption that have one or more insurances (black dots) and two or more insurances (hollow dots).

Remark 2.1 (*Multiple endogenous variables*) Our results also hold for X_i an arbitrary vector which satisfies equation (1) (i.e., instead of $X_i = (1(T_i \geq t_1), \dots, 1(T_i \geq t_d))'$, X_i can be any combination of continuous or discrete variables). We show in Appendix C (see the proof of Proposition B.1) that equation (2) holds under the same assumptions of the standard case. This implies that all other results in this paper also hold unchanged in the general case, since all use equation (2) directly, and do not rely on the specific definition of X_i .

In the general case,

$$\beta_j(w) = E[\xi_{ij}\beta_{ij}|W_i = w, \Delta_{ij} > 0],$$

which is an average of the effects of a 1 unit increase in X_{ij} conditional on $W_i = w$ among compliers for treatment variable j , at the threshold z_0 . Compliers for treatment variable j in this case are observations i such that $\Delta_{ij} > 0$, i.e. all those who would increase X_{ij} when crossing the threshold.

Whenever $\delta_{X_j}(w) > 0$, the weights are defined as $\xi_{ij} = \Delta_{ij}/E[\Delta_{ij}|\Delta_{ij} > 0]$, and so $\xi_{ij} \geq 0$, and $E[\xi_{ij}|\Delta_{ij} > 0] = 1$. Therefore, $\beta_j(w)$ weighs the treatment effects of all individuals proportionally to the increase in X_{ij} across the threshold.⁶

To clarify the meaning of $\beta_j(w)$ in the general case, suppose $X_{i1} = \text{income}$, $X_{i2} = \text{daily number of cigarettes}$, $W_i = \text{race}$, and $w = \text{white}$. Then, equation (1) implies that the outcome is linear in income and cigarettes i.e. no matter the person's original income level, the effect of a 1 dollar increase in income is the same (equal to β_{i1}), and no matter how much the person smokes, the effect of smoking 1 additional daily cigarette is the same (equal to β_{i2}). In this case, $\beta_1(w)$ is an average of the effect of a 1 dollar increase in income among all whites who would increase their income if crossing the threshold. Analogously, $\beta_2(w)$ is an average of the effect of a 1 cigarette increase among all whites who would increase their smoking if crossing the threshold. The weights ξ_{i1} and ξ_{i2} are proportional to the potential increases in income and

⁶One need not be concerned with $\beta_j(w)$ whenever $\delta_j(w) = 0$, because no identifiable quantity discussed in this or any other paper can be a function of $\beta_j(w)$, per Proposition B.4 in Appendix B.2. One may therefore ignore $\beta_j(w)$ entirely in all assumptions without consequences. If desired, for completeness one may define $\beta_j(w) = \beta_j(w')$ for any w' such that $\delta_{X_j}(w) > 0$. Such w' must exist by Assumption 2, otherwise column j in $E[\delta_X(W)\delta_X(W)]$ is equal to zero.

smoking across the threshold, respectively. So, suppose that the expected income increase among those who would increase their income across the threshold is \$4,000 ($\mathbb{E}[\Delta_{i1}|\Delta_{i1} > 0] = 4,000$), and person i 's income would have increased 2,000 dollars across the threshold ($\Delta_{i1}2,000$), then $\xi_{i1} = 2,000/4,000 = 0.5$. Analogously, if the expected increase in cigarettes among those who would increase their smoking across the threshold is 2, and person i would have increased 3 cigarettes, then $\xi_{i2} = 3/2 = 1.5$.

3 Discussion of Assumption 1

Assumption 1 imposes a restriction on $\beta_j(w)$, the vector of average treatment effects. Specifically, it excludes W from influencing the average marginal effects, $\beta_j(w) = \beta_j$. As a comparison, the linearity restriction often imposed in the RDD literature excludes the treatment T from influencing average effects, which translates into $\beta_j(w) = \beta(w)$. In general, to identify multiple treatment effects when the treatment variable is multivalued one must make assumptions that restrict the heterogeneity of the treatment effects. While to date the linearity assumption is the only available restriction in the researcher's toolkit, this paper suggests that a different restriction is also available, which may be more plausible depending on the context, as in the case of our application.

In the following sections, we discuss how Assumption 1 can be justified in an application (Section 3.1), how it may be tested (Section 3.2), or relaxed (Section 3.3). Finally, we analyze the robustness properties of our approach when Assumption 1 is violated (Section 3.4).

3.1 Justifying Assumption 1

Assumption 1 seems to be at odds with Assumption 2. How can it be the case that the treatment effects do not vary with W , while the first stages do? It turns out first stages may be heterogeneous for many reasons other than differences in treatment effects. Assumptions 1 and 2 are both valid, in fact, when the choice of treatment, T , depends on the value of W only through the

marginal cost and some specific pieces of the marginal benefit. Appendix A demonstrates this result in detail with a simple framework where people choose the level of treatment in order to maximize their overall surplus, which depends on costs, benefits related to the outcome variable Y and other benefits unrelated to the outcome variable Y (e.g., Roy 1951, Björklund and Moffitt 1987, Eisenhauer et al. 2015). Here, we discuss the main arguments that are formalized in that appendix.

Suppose that individuals choose T by finding the level that maximizes the benefit subject to the cost of the choice. Then, the chosen level of T is that which equates the marginal benefit and the marginal cost. This framework allows us to relate Assumption 1 to the reasons why people choose different levels of T . Specifically, if W influences the choice of T only through the marginal cost of T , then, since the treatment effect is part of the marginal benefit of T , Assumption 1 holds.

In our application, if individuals value healthcare utilization, which is plausible, and differences in the choice of quantity of insurance, T , across race are due only to differences in cost, then Assumption 1 must hold for $W = \text{race}$. Indeed, the literature that studies racial differences in health insurance coverage seems to be in consensus with regards to the reasons for such differences, in attributing the racial differences in coverage to differences in the cost of acquiring insurance, be it through the access to jobs offering insurance, available disposable income, and types of insurance offerings in their choice set.⁷ See Pylypchuk and Selden (2008) for a comprehensive survey of this literature. Analogously, for $W = \text{education}$, the literature that studies the sources of discrepancies in insurance coverage by education level also seems to attribute differences in coverage to constraints of a similar nature, especially in access to employer-sponsored insurance (through their jobs or their spouse’s), or through more disposable income to purchase insurance some other way. See for instance Farber and Levy (2000).

⁷For instance, Artiga et al. (2021) state that “The higher uninsured rates among people of color reflected more limited access to affordable health coverage options. Although, the majority of individuals have at least one full-time worker in the family across racial and ethnic groups, people of color are more likely to live in low-income families that do not have coverage offered by an employer or to have difficulty affording private coverage when it is available.”

The justification of Assumption 1 offered in the last paragraphs is sufficient but not, in fact, necessary. We may relax it in two ways. First, Assumption 1 holds even if W influences the choice of T through some parts of the marginal benefit as well, as detailed in Appendix A, as long as the choice of T is not influenced by W through variations in treatment effects. In our application, race and education are allowed to influence the choice of number of insurances through insurance's cost, as discussed above, and through (1) the direct marginal benefit of the amount of insurance coverage keeping health care utilization constant (the direct marginal benefit of T keeping Y constant), for example due to different preferences over other goods across race and education; (2) the direct marginal benefit of healthcare utilization keeping the number of insurances constant (the direct marginal benefit of Y keeping T constant), for example due to different propensity to practice moral hazard across race and education; and (3) the level of healthcare utilization, Y , for example due to different baseline health across race and education.

Second, the arguments above must be valid only among compliers of treatment level j , for each j . This weakens the burden of proof, because compliers of a given treatment level j and different W are more comparable than arbitrary individuals with different W . For example, in our application, the arguments above must hold across individuals of different race and education that share enough economic and preference similarities so as to be compliers of the same treatment level.⁸ Thus, although whites may choose to hold more insurances because they expect higher treatment effects due to differences in wealth (e.g. because whites are wealthier, they may expect that, upon obtaining additional insurance, they would increase their healthcare usage proportionally more than minorities, because they can better afford copays), such differences only need to be ruled out for compliers of the same treatment level, who are likely to have similar enough wealth so as to make the same choices before and after qualifying for Medicare.

Finally, this framework clarifies that Assumptions 1 and 2 are not necessarily at odds because

⁸As discussed in Section 2, compliers of treatment level 1 are those who would not be insured if slightly younger than 65, but would be insured if slightly older than 65. Compliers of treatment level 2 are those who would hold one or zero insurances if slightly younger than 65, but would hold two or more insurances if slightly older than 65.

even if Assumption 1 holds – and thus treatment effects do not vary with W – the first stages may vary in W due to changes at the threshold in marginal cost (which is the case in our application), and all parts of the marginal benefit except the treatment effects, as described above.

3.2 Testing Assumption 1

Over-identification can be used to test Assumption 1. Suppose that the homogeneity in covariates restriction is wrong. In this case,

$$\delta_Y(W) = \beta' \delta_X(W) + (\beta(W) - \beta)' \delta_X(W),$$

and thus the moment conditions $E[(\delta_Y(W) - \beta' \delta_X(W))W] = 0$ are not satisfied. This is exactly what over-identification tests such as Sargan-Hansen’s J -test are built to detect. As we will show in Section 4, our estimators can be transformed into standard 2SLS estimators and thus performing the test in packaged software is as simple as running the regression under the assumed model and then selecting the option to display the over-identification test results.

3.3 Relaxing Assumption 1

Over-identification can also be used to relax Assumption 1. Suppose instead the correlated random coefficients model

$$\beta(W) = \theta_0 + \theta_1' \tilde{W}, \tag{4}$$

where $\tilde{W}$ is a vector with c elements which are functions of the elements of W , and may include the elements of W themselves, indicators of sets of values of W , interactions, higher order terms, etc., and may even not include some elements of W at all. The heterogeneous treatment effect $\beta(W)$ is parameterized by $\theta = (\theta_0', \text{vec}(\theta_1)')'$ ($\text{vec}(A)$ transforms the columns of A into a single column vector). For identification we need to impose the restriction that $c \leq q/d - 1$, which limits the dimension of $\tilde{W}$. For instance, $\tilde{W}$ cannot be a fully saturated set of indicators of all

the values in the support of the distribution of W .

For illustration, in our application, if $W = \text{race and education}$, then $q = 6$, and therefore Assumption 1 can be relaxed by assuming instead that $\beta(W) = \theta_0 + \theta_1 \tilde{W}_1 + \theta_2 \tilde{W}_2$ for $\tilde{W}_1$ and $\tilde{W}_2$ any two functions of W . For example, we can suppose $\tilde{W}_1$ to be an indicator that the person is a minority, and $\tilde{W}_2 =$ an indicator that the person is a high school dropout. Then, $\beta(W)$ is allowed to vary freely with race and high school dropout status, but must satisfy the following homogeneity conditions: $\beta(\text{white, high school graduate}) = \beta(\text{white, some college})$ and $\beta(\text{minority, high school graduate}) = \beta(\text{minority, some college})$.

The identification equations can be written as

$$\delta_Y(W) = \theta'[(1, \tilde{W}) \otimes I] \delta_X(W),$$

where $\otimes$ denotes the Kronecker product. Define $\tilde{X} = (X', \tilde{W}_1 X', \dots, \tilde{W}_c X')'$, where $\tilde{W}_1, \dots, \tilde{W}_c$ are the elements of $\tilde{W}$. Then $\delta_Y(W) = \theta' \delta_{\tilde{X}}(W)$, which is equivalent to a model with homogeneous treatment effects in the variable $\tilde{X}$.

All the assumptions and identification results in this paper also hold when X includes variables that are not binary (see Remark 2.1). Therefore, the vector of parameters θ can be identified per equation (3), using $\tilde{X}$ instead of X . Of course, one can both relax Assumption 1 and test the relaxed version by specifying $\beta(W)$ parsimoniously, so θ is over-identified.

3.4 Robustness under violation of Assumption 1

What are the consequences of applying this approach under a failure of Assumption 1? In this case, the identified quantity $\bar{\beta}_W = E[\delta_X(W) \delta_X(W)']^{-1} E[\delta_X(W) \delta_Y(W)']$ is the unique solution to the minimum distance problem

$$\bar{\beta}_W = \arg \min_{\gamma} E[(\beta(W) - \gamma)' \Lambda(W) (\beta(W) - \gamma)], \quad (5)$$

where $\Lambda(W) = \delta_X(W)\delta_X(W)'$. Therefore, if Assumption 1 fails, our approach provides a quantity which minimizes a distance to the true values $\beta(W)$.

One might question whether there may be other weight functions $\Lambda(W)$ which are preferable to $\delta_X(W)\delta_X(W)'$, yielding an approach with better properties than ours. The answer is no. In Appendix B.2, we show that among all weight functions, using $\delta_X(W)\delta_X(W)'$ is the optimal approach, in the sense that it yields the quantity with the strongest identification.

Moreover, $\bar{\beta}_W$ is a convenient weighted average of the effects of interest. Specifically, we show in Appendix B.2 that the j -th coordinate of $\bar{\beta}_W$ satisfies $\bar{\beta}_{Wj} = E[\omega_{jj}(W)\beta_j(W)] + \sum_{s \neq j} E[\omega_{js}(W)\beta_s(W)]$, where the weights satisfy $E[\omega_{jj}(W)] = 1$, and $E[\omega_{js}(W)] = 0$ when $s \neq j$. Therefore, under a failure of Assumption 1, the j -th coordinate of our estimator identifies an average of the $\beta_j(W)$ with weights that average one, which is contaminated by an average of the other LATEs, $\beta_s(W)$ for $s \neq j$, with weights that average zero.

By Cauchy-Schwarz, the contamination term satisfies

$$\left| \sum_{s \neq j} E[\omega_{js}(W)\beta_s(W)] \right| \leq \sum_{s \neq j} SD(\omega_{js}(W)) \cdot SD(\beta_s(W)),$$

where $SD(\cdot)$ is the standard deviation.⁹ This implies that even if $\beta_j(W)$ does not satisfy Assumption 1, our approach is “locally robust,” in the sense that if for all $s \neq j$, the $\beta_s(W)$ do satisfy Assumption 1, then $SD(\beta_s(W)) = 0$, and thus $\bar{\beta}_{Wj}$ is an average of the $\beta_j(W)$ without any contamination from the other marginal treatment effects. If for some $s \neq j$, the $\beta_s(W)$ do not satisfy Assumption 1, but $SD(\beta_s(W))$ is small, then the amount of contamination will be small.

4 Estimation

We propose an estimator that retains the structure as similar as possible to the standard RDD while allowing for the inclusion of covariates and multiple treatments.

⁹The inequality holds because, since $E[\omega_{js}(W)] = 0$, $E[\omega_{js}(W)\beta_s(W)] = E[\omega_{js}(W)(\beta_s(W) - E[\beta_s(W)])]$.

Our estimator is equivalent to the sample version of the identification implied by equation (2). That is, it is equivalent to a two-step process: (1) estimate the reduced form and first-stage discontinuities using local linear regression; and (2) run a local linear regression of the estimated reduced-form discontinuities on the estimated first-stage discontinuities. This estimator can be implemented rather simply as a Two-Stage Least Squares (2SLS) regression, which we describe next.

Consider the following definitions.

1. Recenter Z so that $z_0 = 0$ (subtract z_0 from all Z_i so that the threshold now falls at zero). Define $D = 1(Z \geq 0)$.
2. Define $C = (1, W', Z, D \cdot Z, Z \cdot W', D \cdot Z \cdot W')'$.¹⁰
3. Let $k_{h_n}(Z) = k(Z/h_n)$, where k is a kernel function, and h_n is a bandwidth parameter satisfying some standard conditions in Appendix D.

We formalize our approach as a weighted 2SLS regression with first-stage regressions

$$X_{ij} = \pi_{0j}D_i + \pi'_{1j}W_i \cdot D_i + \alpha'_j C_i + \varepsilon_{ij}^x, \quad j = 1, \dots, d$$

second stage regression

$$Y_i = \beta' \hat{X}_i + \eta' C_i + \varepsilon_i^y,$$

where $\hat{X}_i$ is the predicted outcome of the first stage regression, and the weights are equal to $k_{h_n}(Z_i)$. In other words, we propose running a weighted 2SLS regression of Y on X using D and $D \cdot W$ as IVs, C as exogenous controls, and k_{h_n} as weights.¹¹ Standard errors and

¹⁰Although for identification it only matters how many values W assumes, for estimation we need W to be a vector of at least $d - 1$ elements. If W has a finite support $\{w_1, \dots, w_L\}$, a natural choice is to redefine $W = (\mathbf{1}(W = w_1), \dots, \mathbf{1}(W = w_{L-1}))'$. Nevertheless, any other vector of functions of the original W with at least $d - 1$ elements is acceptable, provided the rank condition of the 2SLS regression is satisfied.

¹¹In Stata, the code is “ivregress 2sls Y C (X = D DW) [w=k]” where the option “[w=k]” leads to a weighted regression with weight equals to the variable k, where different kernels and bandwidths can be chosen. In principle one could use the option “gmm” instead of “2sls” in this Stata command, which in our application yielded very similar results. The study of the properties of this method using a GMM estimator is, however, beyond the scope of this paper.

tests obtained directly from packaged software can be used for inference, under appropriate assumptions discussed below.

Under the parametric random coefficients model discussed in Section 3.3, estimation should be performed identically, substituting X by $\tilde{X}$, and β by θ . Here, $\hat{\theta}_j$ estimates the j -th element of the vector θ .

As mentioned above, although the estimator is implemented as a weighted 2SLS regression, it is actually a local linear regression on the running variable Z for each side of the threshold and for both the first stage and second stage regressions. To see this, note that, similarly to the standard RDD, this estimator weighs observations by the kernels, which restricts the sample only to the proximity of the threshold, and includes in the regression (inside of C) the terms 1, Z and $D \cdot Z$. This allows the expected outcome to vary with the running variable, and also to change derivative across the threshold. However, different from the standard RDD, this estimator includes W , $Z \cdot W$ and $D \cdot Z \cdot W$ in C , which allows the outcome to vary with the running variable differently for every element of W , and also to change slope across the threshold differently for every element of W .

We use the local linear estimator for its excellent boundary properties (see, e.g. Imbens and Lemieux 2008), as does the current RDD literature. Because we use the local linear method on a single running variable Z , and a single threshold, some of what is already known in the RDD literature can be applied directly here.¹² For instance, we know that triangular kernels tend to perform better in boundary estimation (Lee and Lemieux 2010), and we know that the use of controls can improve the efficiency of the estimators (Calonico et al. 2019). Thus, if there are leftover covariates in the data which are not used in W , they could be included as additional terms in the vector C .

The following result establishes the asymptotic normality of the proposed estimator. Its regularity conditions and proof are given in Appendix D.

¹²Accordingly, many valuable extensions to the standard RDD case can be adapted to our setting, such as the use of higher order polynomials, or the consideration of measurement error in the running variable (e.g. Dong 2015 and Bartalotti et al. 2021).

Theorem 4.1 *Let Assumptions 1, 2, 3 (Appendix B) and 4 (Appendix D) hold. Then*

$$\sqrt{nh_n}(\hat{\beta} - \beta) \rightarrow_d N(0, \Sigma),$$

where Σ is given in Appendix D.

Note that if Assumption 1 does not hold, then Theorem 4.1 still holds with $\bar{\beta}_W$ instead of β , per Section 3.4. Note also that Theorem 4.1 can be immediately adapted to the estimator in the parametric random coefficients model $\beta(W) = \theta_0 + \theta_1 \tilde{W}$ (discussed in Section 3.3).¹³

Assumption 4 in Appendix D requires undersmoothing, in the sense that the bandwidth satisfies $nh_n^5 \rightarrow 0$. Without this assumption, inference based on asymptotic normality is invalid. Nevertheless, our estimates are remarkably stable to variations in bandwidth, and we have observed similar stability in other applications as well. A formal study of bandwidth choice as well as robustness of our estimator to bandwidth variation is beyond the scope of this paper. Other valuable extensions left for future research include inference under the optimal bandwidth as well as bias-correction. At present, we recommend that the practitioner consider a few different choices of bandwidth around the optimal bandwidth for undersmoothing from Calonico et al. (2020).

5 The Effect of Insurance Coverage on Health Care Utilization

We apply our approach to the problem of estimating the effects of insurance coverage on health care utilization with a regression discontinuity design, as in Card et al. (2008), henceforth CDM. They exploit the fact that Medicare eligibility varies discontinuously at age 65. Medicare eligibility may affect health care utilization via two channels. First, it provides coverage to

¹³In this case, all assumptions must hold with $\tilde{X}$ instead of X . Define $\tilde{\Sigma}$ identically to Σ , but using $\tilde{X}$ instead of X . Then, the result is $\sqrt{nh_n}(\hat{\theta} - \theta) \rightarrow_d N(0, \tilde{\Sigma})$, and $\tilde{\Sigma}$ can be consistently estimated using the standard 2SLS variance estimator.

people who were previously uninsured. Second, it provides more generous coverage to people who are also insured by other insurance policies. Let Y be a measure of health care use (e.g., whether the person did not get care for cost reasons last year). The two treatment variables are an indicator of whether the person has one or more insurance policies, X_1 , and an indicator of whether the person has two or more insurance policies, X_2 . The running variable Z is the age (measured in quarters of a year) relative to the threshold of 65 years of age. The Medicare eligibility status, D , is an indicator of whether the person is 65 years of age or older. We want to identify LATEs β_1 and β_2 , which are respectively the local (to the threshold and to compliers) average treatment effects of X_1 and X_2 on Y . Note that β_1 is identified among compliers for treatment level 1, while β_2 is identified among compliers for treatment level 2, and these two sets of compliers are likely different (see Section 2).

Table 1 presents the summary statistics for the key variables in our sample, obtained from the National Health Interview Survey (NHIS) from 1999 to 2003. We consider three different outcome variables Y : whether the person delayed care last year for cost reasons, whether the person did not get care last year for cost reasons, and whether the person went to the hospital last year.¹⁴

The table shows that less educated people and minorities are more likely to delay or ration care because of cost reasons. However, high school dropouts tend to go more to the hospital than their more educated counterparts, irrespective of their race. Additionally, non-Hispanic whites tend to carry more insurance policies than minorities with the same education level. Finally, people with more education are more likely to have some insurance policy than people with less education, irrespective of their race, although heterogeneity across education levels is less pronounced than heterogeneity across race.

Interestingly, less educated whites are more likely to carry two or more insurance policies

¹⁴There is another variable used by CDM: whether the person saw a doctor last year. This variable has many missing data throughout the sample period, so we opted to drop it from our analysis. Nevertheless, our estimates for this outcome variable, although less precise, are similar to the ones we obtain for the other low-cost recurrent outcome variables (“delayed care” and “rationed care”).

Table 1: Summary Statistics

Variable	All	Non-Hispanic White			Minority		
		HS Dropout	HS Graduate	Some College	HS Dropout	HS Graduate	Some College
Delayed Care (Y)	0.07 (0.25)	0.09 (0.29)	0.06 (0.24)	0.05 (0.23)	0.10 (0.30)	0.07 (0.25)	0.07 (0.25)
Rationed Care (Y)	0.05 (0.22)	0.07 (0.25)	0.04 (0.20)	0.03 (0.18)	0.09 (0.29)	0.06 (0.23)	0.05 (0.22)
Went to Hospital (Y)	0.13 (0.34)	0.17 (0.38)	0.12 (0.33)	0.12 (0.32)	0.14 (0.35)	0.12 (0.33)	0.12 (0.32)
1+ Coverage (X_1)	0.92 (0.27)	0.91 (0.28)	0.95 (0.22)	0.96 (0.19)	0.78 (0.41)	0.87 (0.33)	0.91 (0.29)
2+ Coverage (X_2)	0.33 (0.47)	0.44 (0.50)	0.38 (0.49)	0.32 (0.47)	0.24 (0.43)	0.23 (0.42)	0.24 (0.42)
Medicare Eligible (D)	0.42 (0.49)	0.55 (0.50)	0.45 (0.50)	0.37 (0.48)	0.46 (0.50)	0.36 (0.48)	0.34 (0.47)
Observations	63,165	8,337	16,037	21,352	8,293	4,302	4,844

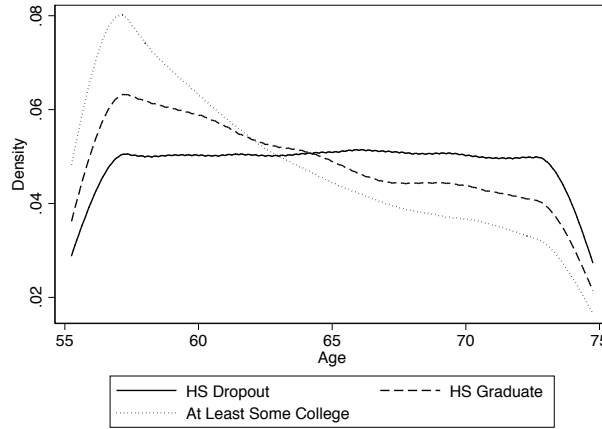
Note: Source: NHIS 1999-2003. Standard deviations in parentheses. “HS Dropout” represents people who have less than a high school degree. “HS Graduate” represents those who have exactly a high school degree. “Some College” represents those who have at least some college. This sample reflects only people who are between 55 and 75 years old.

than their more educated counterparts. This counter-intuitive correlation is better understood in the context of age as an important confounder. As seen in the second to last row of the table, people with more education are less likely to be eligible to Medicare. Indeed, Figure 4 shows that people with less education in the sample tend to be older than people with more education, so they are mechanically more likely to be eligible to Medicare.

Following CDM, we use a regression discontinuity design to circumvent this and other endogeneity concerns. To see how an RDD is reasonable in this context, Figure 5 shows that people just younger and just older than 65 years of age are comparable in terms of race and education. Similarly to CDM, we find no evidence of discontinuity at 65 years of age for a wide range of covariates.¹⁵

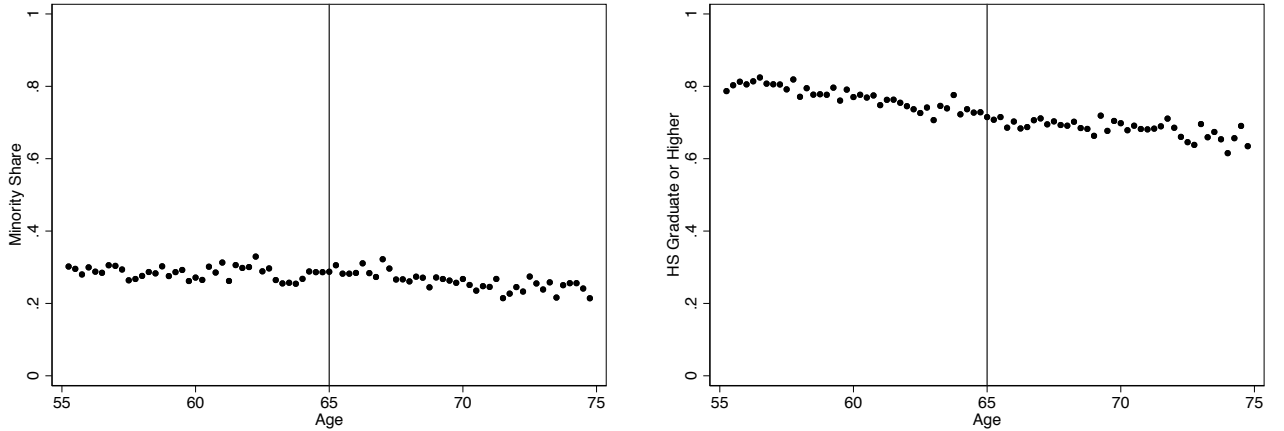
¹⁵Some of the potential confounders analyzed are the person’s income, work status and health status. See Section II.A in CDM for a more detailed discussion of potential confounders in this context.

Figure 4: Age Distribution by Level of Education



Note: For each education level, this figure shows the kernel density plot of the age distribution (measured in quarters of a year), using the Epanechnikov kernel and bandwidth equal to 1.

Figure 5: Validity Plots



Note: The scatter plot shows the average of the variable described in the vertical axis for each age level (measured in quarters of a year).

The covariates available in the data include indicators for education (high school dropouts, high school graduates, and those with at least some college), race (non-Hispanic whites and minorities), gender, Hispanic ethnicity, region of the country (four indicators) and year of the sample (five indicators). Of these covariates, we must choose which ones to include in W . The other variables not included in W may be ignored, as in the standard RDD, or may be included as standard controls (i.e. C in Section 4) to improve efficiency, as in the standard RDD case with covariates.

To keep the application as close to CDM as possible, we report results based on the same choices as in that paper, but in all cases we check the robustness under alternative choices. Thus, we (1) include the additional covariates not included in W as elements of C (therefore, this application mirrors more the standard RDD with covariates than the standard RDD);¹⁶ (2) report the results with bandwidth $h = 10$ and uniform kernel;¹⁷ and (3) cluster standard errors by Z .¹⁸

Table 2 shows the first stage and reduced-form estimates for each subsample defined by all possible breakdowns of race and education. The first panel shows the first stage estimates (i.e.

Table 2: First Stage and Reduced Form Estimates: Standard RDD

First Stage	All	Dropout	Whites HS Grad	College	Dropout	Minorities HS Grad	College
X_1 : 1+	0.093	0.136	0.069	0.048	0.209	0.100	0.075
Coverage	(0.004)	(0.014)	(0.005)	(0.004)	(0.016)	(0.019)	(0.014)
X_2 : 2+	0.482	0.460	0.584	0.589	0.227	0.322	0.376
Coverage	(0.018)	(0.022)	(0.025)	(0.025)	(0.016)	(0.034)	(0.025)
Reduced Form (Y)	All	Dropout	Whites HS Grad	College	Dropout	Minorities HS Grad	College
Delayed Care	-0.018	-0.008	-0.014	-0.012	-0.053	-0.027	-0.001
	(0.004)	(0.012)	(0.007)	(0.006)	(0.010)	(0.012)	(0.014)
Rationed Care	-0.016	-0.004	-0.019	-0.012	-0.038	-0.022	0.004
	(0.003)	(0.012)	(0.006)	(0.004)	(0.010)	(0.012)	(0.011)
Went to Hospital	0.010	0.006	0.020	0.015	-0.016	0.000	0.019
	(0.005)	(0.016)	(0.010)	(0.008)	(0.016)	(0.016)	(0.018)

Note: This table shows the first stage (first panel) and reduced-form effects (second panel) for each variable defined in the first column, and each subsample referred by the remaining columns. “Dropout” represents people who have less than a high school degree. “HS Grad” represents those who have exactly a high school degree. “College” represents those who have at least some college. These estimates were obtained from a local linear regression with uniform kernel and bandwidth $h = 10$ ($N=63,165$), with standard errors clustered by Z .

¹⁶The reported estimates are almost identical to the ones where we do not include these covariates in C .

¹⁷We also implemented the approach with triangular kernel, and with bandwidths ranging from 5 to 10, obtaining very similar results.

¹⁸The clustered standard errors turned out to be very similar to the unclustered standard errors in this context.

the effect of Medicare eligibility on X_1 and X_2), while the second panel shows the reduced-form estimates (i.e. the effect of Medicare eligibility on the outcome Y). We start by noting that the aggregated reduced-form results (lower panel, second column) suggest that there is a significant discontinuity in each of the three outcome variables. However, is this evidence of an effect of which treatment variable: X_1 (“1+ Coverage”) or X_2 (“2+ Coverage”)? It is not possible to answer this question using the standard RDD framework, since both treatment variables vary discontinuously at the threshold: while Medicare eligibility increases the number of people covered by one insurance policy by about 9%, it also increases the number of people covered by two policies by about 48% (upper panel, second column).

In the remaining columns of Table 2, we show the discontinuity estimates broken down by race and education. It is clear that there is a first stage discontinuity in both treatment variables X_1 and X_2 for each and every combination of race and education (this can also be seen in Figure 3 in Section 2, which shows the first stage plots for each combination of race and education levels). Thus, we cannot find one specific combination of race and education where only one of these treatment variables vary discontinuously, which would allow us to isolate the effect of one of the treatment variables (by Proposition B.2 in Appendix B.1).

In Table 3, we report the main estimates of the paper using our proposed estimator (Section 4) for different outcome variables Y and different choices of W . Under Assumption 1 ($\beta_1(W) = \beta_1$ and $\beta_2(W) = \beta_2$), the estimates are interpreted as the LATEs of X_1 and X_2 . When there are over-identifying restrictions, we test these exclusion restrictions with an over-identification test (the Sargan–Hansen’s J -test), and the p-value is shown in brackets for each outcome Y and each choice of W .

We start from the most parsimonious specification of W . In column (I), we report the results for W equal to an indicator of race (i.e., indicator equals one if minority, and zero if non-Hispanic white). In this scenario we are just identified, so we are unable to test the exclusion restriction $\beta(\text{Race}) = \beta$. We find β_1 to be negative for “Delayed Care” and “Ration Care,” and negative (but insignificant at standard levels) for “Went to Hospital.” In contrast, we find that β_2 is close

Table 3: Main Results

Outcome	(I) $W = \text{Race}$		(II) $W = \text{Race \& Educ}$		(III) $W = \text{Race, Educ \& Other}$	
	$\hat{\beta}_1$	$\hat{\beta}_2$	$\hat{\beta}_1$	$\hat{\beta}_2$	$\hat{\beta}_1$	$\hat{\beta}_2$
Delayed Care	-0.234	0.008	-0.210	0.004	-0.203	0.001
	(0.060)	(0.013)	(0.050)	(0.010)	(0.043)	(0.009)
			[J-test p-value = 0.360]		[J-test p-value = 0.347]	
Rationed Care	-0.145	-0.005	-0.132	-0.007	-0.135	-0.007
	(0.053)	(0.009)	(0.043)	(0.007)	(0.038)	(0.007)
			[J-test p-value = 0.458]		[J-test p-value = 0.133]	
Went to Hospital	-0.087	0.038	-0.109	0.041	-0.084	0.035
	(0.086)	(0.018)	(0.081)	(0.017)	(0.074)	(0.016)
			[J-test p-value = 0.925]		[J-test p-value = 0.516]	

Note: This table shows, for each horizontal panel represented by a different outcome variable Y , 2SLS estimates of β_1 and β_2 applying our multivariate RDD approach. In column (I), W is an indicator for race (i.e., minority status). In column (II), we use indicators of the six groups formed by combinations of race and education. In column (III), we add indicators for gender, Hispanic ethnicity, region and year of the sample. In the three sets of columns, all covariates available in the sample but not included in W are included in C (see Section 4 for a description of C) for efficiency, following CDM. Estimates are interpreted under the exclusion restriction that $\beta_1(W)$ and $\beta_2(W)$ do not change with W . Whenever possible, we also show the p-value of the test of this exclusion restriction via the Sargan-Hansen’s J -test. Estimates were obtained from a local linear regression with uniform kernel and bandwidth $h = 10$ ($N=63,165$), with standard errors clustered by Z .

to zero for “Delayed Care” and “Ration Care,” and positive for “Went to Hospital.”

In column (II) we make W more complex, including indicators of all combinations of race (i.e., minority indicator) and education (whether the person is a high-school dropout, a high-school graduate, or has at least some college education). The implied exclusion restriction is stronger than in column (I), because it includes education as well as race ($\beta(\text{Race, Education}) = \beta$), but now we have four over-identifying restrictions to test this assumption. The point estimates in all cases are very similar and are statistically the same as the ones from column (I). The standard errors are also uniformly smaller, now that we can exploit further heterogeneity in the first stages. The p-values from the J-tests suggest no evidence that the LATEs vary by either race or education, or both.

Finally, in column (III) we expand W further by adding all the other remaining covariates in our dataset: female, Hispanic ethnicity, region and year indicators. This greatly strengthens the exclusion requirement in Assumption 1, by requiring that the LATEs do not vary with these variables as well. At the same time, we provide more precise estimates, as we exploit further heterogeneity in the first stages. More importantly, now we have fourteen over-identifying restrictions to test the exclusion restrictions. As the p-values of the J-tests show, we now find some evidence that these exclusion restrictions might be invalid for “rationed care.” Because of this, the results from columns (I) and (II) are our preferred ones, since there the exclusion restrictions seem to be valid for all outcome variables. Nevertheless, it is interesting to note that the point estimates from column (III) are remarkably similar to the ones from column (II), suggesting that this over-identification test might have power to detect violations from the exclusion restrictions which do not yet bias our estimates too much. This speaks to the robustness of our estimator to small violations from Assumption 1, as discussed in Section 3.4.

Overall, we find that Medicare as the only insurance causes a reduction of about 15 percentage points in the delay or ration of health care for cost reasons, and Medicare as a second insurance causes an increase of about 4 percentage points in hospital visits. Our results confirm the intuition suggested by CDM that Medicare eligibility has generated different effects on health care utilization depending on whether the person had insurance before becoming eligible to Medicare. Those with originally no insurance are now able to not delay or to not ration their care. We also find some weak evidence that these people went less frequently to hospitals, although these estimates are not significant at standard levels. One possible interpretation is that the increase in low-cost care (see β_1 estimates in Table 3 for the outcome variables “delayed care” and “rationed care”) may have generated some positive effects in the prevention of more serious health problems, yielding some weak evidence of less frequent hospital visits. Investigating this further would require more detailed data, and is beyond the scope of this paper. At the same time, those with insurance beyond Medicare may have been affected by Medicare along a different margin. They do not seem to have changed at all their health care

utilization for low-cost, recurrent cases (see β_2 estimates in Table 3 for the outcome variables “delayed care” and “rationed care”), but they have increased their hospital visits, which tend to be more costly. With more detailed data, it would be valuable to further investigate the actual reason why these people are going to the hospital more frequently.

Remark 5.1 *We consider an exercise that is illustrative of the limitations of ad hoc strategies using the standard RDD in such a setting. We analyze two strategies: (A_1) implementing the standard RDD approach with the indicator X_1 as the only treatment variable; (B_1) implementing strategy A_1 , but first restricting the sample to individuals with $X_{2i} = 0$ (i.e., restricting to individuals with either 0 or 1 insurance). Analogously, one might want to consider similar strategies involving X_2 : (A_2) implementing the standard RDD approach with X_2 as the only treatment variable; (B_2) implementing strategy A_2 , but first restricting the sample to individuals with $X_1 = 1$ (i.e., restricting to individuals with at least 1 insurance).*

We know theoretically that these strategies are not appropriate if the objective is the identification of local average treatment effects at the extensive and intensive margins, because both methods fail the assumptions of the standard RDD. Strategy A_1 fails the requirement that all other variables that influence the outcome must be continuous across the threshold. As shown in Figure 1, X_2 varies discontinuously at the threshold, and thus the only way we could identify the effect of X_1 is if X_2 does not influence the outcome, i.e. the intensive margin effect is zero. Strategy B_1 introduces a sample selection issue: compliers for treatment level 2 just above the age threshold are not in the sample (since they hold at least two insurances), but the same compliers just below the age threshold are in the sample (since they hold at most one insurance). Analogous arguments are true for strategies A_2 and B_2 .

In any case, Table 4 reports the estimates under these strategies. As expected, irrespective of the outcome variable, the estimates of the coefficients of X_1 for strategy A_1 and strategy B_1 are very different from each other. A comparison with the estimates in Table 3 suggests that for strategy A_1 , the intensive margin effect contaminates the estimated effect of X_1 . The differences are larger for the third outcome (“Went to Hospital”), since the effect of X_2 in that case is likely

farther from zero (this can be inferred by the results from Table 3), so the exclusion restriction $\beta_2 = 0$ implied by strategy A_1 is violated to a greater degree for that outcome. Similar patterns are seen in the results using strategies A_2 and B_2 .

Table 4: Two Naive Strategies Using the Standard RDD

Outcome	Strategy A_1 X_1	Strategy B_1 X_1	Strategy A_2 X_2	Strategy B_2 X_2
Delayed Care	-0.192 (0.035)	-0.001 (0.063)	-0.038 (0.007)	0.006 (0.007)
Rationed Care	-0.173 (0.031)	-0.023 (0.043)	-0.034 (0.006)	0.001 (0.006)
Went to Hospital	0.109 (0.054)	0.059 (0.085)	0.021 (0.011)	0.010 (0.011)

Note: This table shows the limitations of the standard RDD to uncover different marginal effects in the context of this application. The strategies are described in detail in the text.

6 Conclusion

This paper develops a method to identify treatment effects in regression discontinuity designs when the treatment variable takes multiple values. We show that if the treatment effects are homogeneous on a set of covariates, the heterogeneity in the first stages across different values of those covariates can be used to identify, in fact often over-identify, the treatment effects. The over-identification can be used to relax the homogeneity assumption, test it, or both. If homogeneity fails, the approach is robust, in that it identifies a best approximation of the LATEs. The resulting approximation is a weighted average of the LATEs of interest, with weights that add up to one, contaminated by a weighted average of the remaining LATEs, with weights that add up to zero.

We provide an estimator that can be programmed as a weighted Two Stage Least Squares regression on packaged software. The standard errors and test results obtained from packaged

software can be used for inference when the smoothing bias is small.

We apply our method to the estimation of the effect of Medicare health insurance coverage for someone who was not previously insured separately from the effect of adding Medicare coverage for someone who already has another insurance. Our results show that having Medicare as the only insurance increases demand for health care, in the sense that it allows people to not delay or ration health care for costs reasons. However, going to the hospital is determined more by the intensity of coverage, so that having Medicare plus another insurance makes it more likely that someone would seek such expensive type of care.

Many potentially fruitful avenues of methodological research remain open in this setting, so here we mention a few. The formal study of bandwidth choice and bias-correction inference is a valuable endeavour, and so is the study of the robustness of the estimator to variation in bandwidth which we observe in empirical work. An extension of the method to continuous treatment variables is possible, along the lines of [Caetano and Escanciano \(2021\)](#). Finally, it would be valuable to investigate the efficiency gains that can be achieved with over-identification, as well as the study of the power of an over-identification test in this context.

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ONLINE APPENDIX

This online appendix to the paper “Regression Discontinuity Design with Multivalued Treatments” is organized as follows. First, it discusses how Assumption 1 can be justified when the treatment variable is chosen through an optimization problem (Section A). Next, it presents a formal treatment of the identification results in that paper (Section B), as well as proofs (Section C) and asymptotic theory (Section D).

A Justifying Assumption 1 when the Treatment is an Optimal Choice

We examine how Assumption 1 can be justified when the treatment T_i is chosen through an optimization problem. In order to simplify the discussion, we assume here that T_i is continuous, but the same ideas hold when T_i is discrete.

Suppose that the individual’s objective function is $G_i(T_i, Y_i(T_i))$, and the cost of the choice is $C_i(T_i)$.¹⁹ Therefore,

$$T_i = \underset{t}{\operatorname{argmax}} \quad G_i(t, Y_i(t))$$

$$s.t. \quad C_i(t) = 0.$$

Let $G_i(\cdot, Y_i(\cdot))$ be strictly concave and $C_i(\cdot)$ be strictly convex, both differentiable. Denote $G_i^{(t)}(t, Y_i) = \frac{\partial G_i(t, Y_i)}{\partial t}$, $G_i^{(y)}(T_i, y) = \frac{\partial G_i(T_i, y)}{\partial y}$, $Y_i'(t) = \frac{dY_i(t)}{dt}$, and $C_i'(t) = \frac{dC_i(t)}{dt}$. The first order conditions of this maximization problem yield

$$G_i^{(t)}(T_i, Y_i(T_i)) + G_i^{(y)}(T_i, Y_i(T_i)) \cdot Y_i'(T_i) = \lambda C_i'(T_i) \quad (6)$$

where λ is the Lagrangian multiplier.

Consider first the case where individuals choose T_i so as to maximize the outcome Y_i , thus $G_i(t, y) = y$. Here, T_i is chosen through the equation

$$Y_i'(T_i) = \lambda C_i'(T_i).$$

The choice of T_i is influenced by W_i through the marginal cost if $C_i'(T_i) \not\perp W_i$, and through the marginal benefit if $Y_i'(T_i) \not\perp W_i$, where $\perp$ and $\not\perp$ symbolize probability independence and dependence, respectively.

¹⁹Note that uncertainty and private information are allowed here, as the functions $G_i(\cdot)$, $Y_i(\cdot)$ and $C_i(\cdot)$ are indexed by individual.

Define a complier for treatment level t as an observation such that $\lim_{e \downarrow 0} T_i(z_0 - e) < t \leq \lim_{e \downarrow 0} T_i(z_0 + e) > 0$. Define the variable $B(t)$ the set of observations that are compliers for treatment level t , and the average treatment effect function $\beta(t, w) = E[Y'_i(t) | W_i = w, B(t)]$. If

$$Y'_i(t) \perp W_i | B(t), \quad (7)$$

then $\beta(t, w) = \beta(t)$. If the same holds for all t , then Assumption 1 holds. In other words, if individuals choose T_i to maximize $Y_i(T_i)$ and, for each potential treatment level, the choice of T_i among the compliers for that treatment level is influenced by W_i only through the marginal cost of the choice, then Assumption 1 holds.

In general, T_i is chosen through equation (6), and thus we may allow W_i to influence some pieces of the marginal benefit. Specifically, $C_i(t)$, $G_i^{(t)}(t, y)$, $G_i^{(y)}(t, y)$, and even $Y_i(t)$ may be dependent on W_i (so W can be endogenous). The statement above may therefore be relaxed: if individuals choose T_i to maximize an objective function that may depend on Y_i , and for each potential treatment level, the choice of T_i among the compliers for that treatment level is influenced by W_i only through the marginal cost, $C'_i(\cdot)$, the direct marginal benefit of T_i , $G_i^{(t)}(\cdot)$, the direct marginal utility of Y_i , $G_i^{(y)}(\cdot)$, or the outcome level, $Y_i(\cdot)$, but not the marginal return to the outcome $Y'_i(\cdot)$, then Assumption 1 holds.

This model also clarifies how it is possible for there to be heterogeneity in first-stages (Assumption 2), but no heterogeneity in treatment effects (Assumption 1). From equation (6), Assumption 1 can hold, and still the first stages may differ across W because the discontinuities in the distributions of $C'_i(\cdot)$, $G_i^{(t)}(\cdot, \cdot)$, $G_i^{(y)}(\cdot, \cdot)$ or $Y_i(\cdot)$ at the threshold depend on W_i .

B Formal Treatment of the Identification Results

This section presents a formal discussion of identification in an RDD model with multiple treatments in the presence of covariates. We show that identification of marginal treatment effects or interesting weighted averages of the marginal effect of a specific treatment is generally impossible. Then, we show that a convenient weighted average of the marginal effects of all treatments which minimizes the distance to the true effects and has optimality properties is identifiable. Under homogeneity on covariates, or linearity on covariates, identification (in fact over-identification) is possible. Proofs of propositions are in Appendix C.

B.1 Non-identifiability of multivalued LATEs

The following assumption generalizes the conditions of the standard RDD to allow for a multivalued treatment and the presence of covariates. Define $\Delta_{ij}(e) = X_{ij}(z_0 + e) - X_{ij}(z_0 - e)$.

Assumption 3 (i) $E[Y_i(0)|Z_i = z, W_i = w]$ is continuous in z at z_0 almost surely (a.s.) in $w \in \mathcal{W}$; for each j : (ii) $(\beta_i, X_{ij}(z))$ is independent of Z_i near z_0 conditionally on W_i ; (iii) there exists $\varepsilon > 0$ such that for all $0 < e < \varepsilon$: $\Delta_{ij}(e) \geq 0$ a.s. conditionally on W_i ; and (iv) $\Delta_{ij}(e)$ converges almost surely as $e \downarrow 0$, and for all $c > 0$ in the support of $\Delta_{ij}(e)$, $\beta(w, c) := E[\beta_{ij}|W_i = w, \Delta_{ij}(e) = c]$ is continuous in c , and $|\beta(w, c)| \leq g_j(w)$, where $E[g_j(W)] < \infty$.

Assumption 3 is the immediate extension of the assumptions in Hahn et al. (2001) to the multivalued treatment case. We could replace the local independence in Assumption 3(ii) by local smoothness conditions (as in Dong (2018)). Assumption 3(iv) guarantees that $\beta_j(w)$ is well defined.

Proposition B.1 Under Assumption 3, $\beta(w)$ is not identifiable.

This impossibility is not due to the inclusion of the covariates, it is due to the multiple treatments. In particular, if W is constant, this theorem implies the impossibility of identification of the unconditional LATEs $\beta_j = \lim_{e \downarrow 0} E[\beta_j|\Delta_j(e) = 1]$ unless further assumptions are made.

One setting where identification of a causal parameter holds is when, for some group characterized by a particular covariate value w , all but one first stage coefficients are zero. The following proposition states that this case is, in fact, the only circumstance in which identification of LATEs can be achieved, unless further assumptions are made. For any vector a , let a_{-j} denote a without the j -th coordinate.

Proposition B.2 Under Assumption 3, $\beta_j(w)$ is identified $\iff \delta_{X_{-j}}(w) = 0$ and $\delta_{X_j}(w) \neq 0$. In this case, $\beta_j(w) = \delta_Y(w)/\delta_{X_j}(w)$, while $\beta_{-j}(w)$ is completely unidentified.

The idea is that any discontinuity in the outcome $\delta_Y(w)$ must be due only to the discontinuity in X_j , since all the other treatments are continuous across the threshold. Therefore, we can identify the conditional LATE of X_j , $\beta_j(w)$. Proposition B.2 also implies that if $\beta_j(w)$ is identified, then $\beta_s(w)$ is not identified for all $s \neq j$. Without further assumptions, in the best case scenario only one of the marginal effects could be identified for each w .

The restriction $\delta_{X_{-j}}(w) = 0$ may work for some w on a specific application, but it is unlikely to happen in general. Indeed, Figure 3 shows that this “subsetting” strategy cannot yield identification in our application for any combination of race and education, as both variables X_1 and X_2 are discontinuous in all six panels.

Next, we consider the identification of weighted local average treatment effects (WLATE), $E[\omega_j(W)\beta_j(W)]$, for non-trivial identified scalar weights ω_j . Although this quantity can only inform us about a specific average of LATEs across different values of W , it at least includes only marginal effects of X_j , $\beta_j(W)$, avoiding contamination from the marginal effects of X_s ,

$\beta_s(W)$, for $s \neq j$. Unfortunately, the next result shows that the WLATEs can only be identified under the same circumstances in which Proposition B.2 can also be used, and thus these types of averages are not any easier to identify than the LATEs themselves.

Proposition B.3 *Under Assumption 3, $E[\omega_j(W)\beta_j(W)]$ is identified $\iff \omega_j(W)\delta_{X_{-j}}(W) = 0$ a.s.*

This proposition implies that $\omega_j(w)$ can be different from zero only when $\delta_{X_{-j}}(w) = 0$. This means that the identified WLATE $E[\omega_j(W)\beta_j(W)]$ must be a linear combination only of the $\beta_j(w)$ such that $\delta_{X_{-j}}(w) = 0$. However, when $\delta_{X_{-j}}(w) = 0$, Proposition B.2 guarantees that $\beta_j(w)$ is identified. Therefore, Proposition B.3 does not identify any new object that we could not identify with Proposition B.2 already.

Proposition B.3 can thus be seen as an impossibility result: if one cannot apply Proposition B.2 and thus identify $\beta_j(W)$ for some value of W (which may be constant), then one cannot identify any WLATE either.

B.2 Identification of weighted averages of the LATEs

Although it may not be possible to identify the LATEs $\beta_j(W)$, or even the WLATEs $E[\omega_j(W)\beta_j(W)]$ for useful weights, it is possible to identify weighted averages $E[\omega(W)'\beta(W)]$, that combine the LATEs of all X_j , for non-trivial, identified multivariate weights $\omega(W) = (\omega_1(W), \dots, \omega_d(W))'$. We call these TWLATEs, where the T stands for “total,” and denote them by

$$\bar{\beta}_W^\omega = E[\omega(W)'\beta(W)]. \quad (8)$$

Proposition B.4 next gives necessary and sufficient conditions for the identification of these objects. It implies that the only TWLATEs that can be identified are those with weights that are proportional to the first stages.

Proposition B.4 *Under Assumption 3, $\bar{\beta}_W^\omega$ is identified $\iff \omega_j(W) = a(W)\delta_{X_j}(W)$ a.s. for some $a(W)$ and all $j = 1, \dots, d$.*

We are interested in identifying an optimal TWLATE by minimizing the distance to the multivariate conditional LATEs. Define, for a positive semi-definite matrix $\Lambda(W)$,

$$\bar{\beta}_W(\Lambda) = \arg \min_{\gamma} E[(\beta(W) - \gamma)' \Lambda(W) (\beta(W) - \gamma)]. \quad (9)$$

Provided the matrix $E[\Lambda(W)]$ is non-singular, a solution to the minimum distance problem is given by

$$\bar{\beta}_W(\Lambda) = (E[\Lambda(W)])^{-1} E[\Lambda(W)\beta(W)]. \quad (10)$$

According to Proposition B.4, the second factor in (10) is identified if, and only if, $\Lambda(w) = a(w)\delta_X(w)'$, for a vector $a(w)$ of measurable functions. If Assumption 2 holds, the only choice of $a(w)$ that guarantees that $\Lambda(w)$ is positive semi-definite is $a(w) = c(W) \cdot \delta_X(w)$, for a scalar positive function $c(W)$.²⁰ This means that the only optimal TWLATEs that are identifiable are of the form $\bar{\beta}_W(c) = (E[c(W)\delta_X(W)\delta_X(W)'])^{-1}E[c(W)\delta_X(W)\delta_X(W)'\beta(W)]$, where without loss of generality we suppose $E[c(W)^2] = 1$. Our approach is focused on the case where $c(W) = 1$, which gives the quantity in equation (3). The choice of $c(W) = 1$ is optimal, in the sense that $\bar{\beta}_W$ has the strongest identification among all identifiable quantities, because the matrix $E[\delta_X(W)\delta_X(W)'] - E[c(W)\delta_X(W)\delta_X(W)']$ is always positive semi-definite.²¹

Note that if the support of W is finite then Assumption 2 holds if and only if there are d values of W for which the d vectors $\delta_X(W)$ are linearly independent.²² This condition immediately rules out a constant W , because in that case $E[\delta_X(W)\delta_X(W)'] = \delta_X\delta_X'$ (the product of the unconditional first stages) which is a $d \times d$ matrix resulting from the product of a $d \times 1$ vector with another $1 \times d$ vector, and thus can have a rank of at most one. In fact, if W assumes a finite number of values, q , this assumption rules out all W which assume less than d values, so $q \geq d$.²³

An interesting property of $\bar{\beta}_W$ is that it “separates in expectation” the marginal treatment effects of all LATEs, in the sense that the weights average one for the LATEs of one treatment value, and average zero for the LATEs of all the other treatment values. That is, $\bar{\beta}_W = E[\omega(W)\beta(W)]$ for a matrix-valued function $\omega(w)$ with (i, j) -th element $\omega_{jk}(w)$ such that

²⁰Proof: if $\Lambda(W)$ is positive semi-definite, for any vector v with J elements, $v'\Lambda(W)v = v'(\Lambda(W) + \Lambda(W)')v/2$. So, without loss of generality, we can suppose that $\Lambda(W)$ is symmetric in (9). By Proposition B.4, $\Lambda(W) = a(W)\delta(W)'$, which implies that $a(W) = (a_1(W)/\delta_{X_1}(W)) \cdot \delta_X(W)$. (We suppose here that $\delta_{X_1}(W) > 0$. If that is not the case, the result can be derived identically using any other coordinate with a positive first-stage, e.g. $a(W) = (a_2(W)/\delta_{X_2}(W)) \cdot \delta_X(W)$. Assumption 2 guarantees the existence of such coordinate.) Since $\Lambda(W)$ is positive definite, it also follows that $a_1(W)$ must be positive. Define $c(W) = a_1(W)/\delta_{X_1}(W)$, which completes the proof.

²¹Proof: for any vector v with j elements, $A := v'E[(c(W) - 1)\delta_X(W)\delta_X(W)']v = E[(c(W) - 1)\varphi(W)]$, where $\varphi(W) = (v'\delta_X(W))^2$. We can see A as the inner product of $c(W) - 1$ and $\varphi(W)$ in a functional Hilbert space with an L^2 norm. Let $P_a\phi = \varphi(W)(E[\varphi(W)^2])^{-1}E[\varphi(W)\phi(W)]$ be the orthogonal projection of ϕ onto the space generated by φ . If $A \geq 0$, then $A = \|P_a(c(w) - 1)\|$. The norm of the projection P_a is maximized when $\phi = b \cdot \varphi$, for some non-negative constant b . This means that A is largest when $c(W) - 1 = b \cdot \varphi(W)$. Since $E[c(W)^2] = 1$ and $\varphi(w) \geq 0$ is strictly positive with positive probability, it follows that $b = 0$, and thus A is maximized when $c(W) = 1$.

²²Proof: the sum of a positive definite matrix and a positive semi-definite matrix is always positive definite (suppose that A is positive definite and B is positive semi-definite, then for all $v \neq 0$, $v'(A + B)v \geq 0$, and suppose that $v'(A + B)v = 0$, then B would be negative definite, which is absurd.) Say that the $\delta_X(w_l)$ are linearly independent for $l = 1, \dots, d$, then $A = \sum_{l=1}^d \delta_X(w_l)\delta_X(w_l)'P(W = w_l)$ is positive definite. Since $B = \sum_{l=d+1}^q \delta_X(w_l)\delta_X(w_l)'P(W = w_l)$ is positive semi-definite, the result follows.

²³Proof: this follows from the fact that the rank of two matrices is subadditive. If W assumes q values, then $E[\delta_X\delta_X'] = \sum_{l=1}^q \delta_X(w_l)\delta_X(w_l)'P(W = w_l)$. For a given w_l , the rank of $\delta_X(w_l)\delta_X(w_l)'$ is at most 1 (because $\text{rank}(AA') = \text{rank}(A)$.) Therefore, if W assumes q values, the rank of the sum must be at most equal to q , and if $q < d$, the expectation will not be full-rank.

$E[\omega_{jj}(W)] = 1$ and $E[\omega_{jk}(W)] = 0$ if $j \neq k$.²⁴

When the LATEs are homogenous in covariates (Assumption 1), it is possible to identify the β_j . The following result follows immediately from the definition of the minimum distance approximation.

Proposition B.5 *Let Assumptions 2 and 3 hold. If Assumption 1 holds (i.e. $\beta(w) = \beta$), then $\bar{\beta}_W = \beta$.*

C Proofs

Proof of Proposition B.1: We show the result in the general case, where X_i is arbitrary, as in Remark 2.1. We discuss the special case $X_i = (1(T_i \geq t_1), \dots, 1(T_i \geq t_d))'$ in the end. From equation (1) and Assumption 3,

$$\delta_Y(w) = \delta_{\beta'X}(w), \quad \text{for each } w \in \mathcal{W}.$$

However, for small enough $e > 0$, by local independence and monotonicity, for each j ,

$$\begin{aligned} & E[\beta_{ij}X_{ij}|W_i = w, Z_i = z_0 + e] - E[\beta_{ij}X_{ij}|W_i = w, Z_i = z_0 - e] \\ &= E[\beta_{ij}\{X_{ij}(z_0 + e) - X_{ij}(z_0 - e)\}|W_i = w] = E[\beta(w, \Delta_{ij}(e))\Delta_{ij}(e)|W_i = w] \\ &\rightarrow_{e \downarrow 0} E[\beta_{ij}\Delta_{ij}|W_i = w] = E[\beta_{ij}\Delta_{ij}|W_i = w, \Delta_{ij} > 0]P(\Delta_{ij} > 0|W_i = w), \end{aligned} \quad (11)$$

where the limit follows by the Continuous Mapping Theorem and the Dominated Convergence Theorem. If $\delta_j(w) > 0$, then $P(\Delta_{ij} > 0|W_i = w) > 0$, and $E[\Delta_{ij}|\Delta_{ij} > 0] > 0$. Therefore, we can write

$$\begin{aligned} (11) &= E \left[\beta_{ij} \frac{\Delta_{ij}}{E[\Delta_{ij}|W_i = w, \Delta_{ij} > 0]} \middle| W_i = w, \Delta_{ij} > 0 \right] E[\Delta_{ij}|W_i = w] \\ &= \beta_j(w)\delta_j(w), \end{aligned} \quad (12)$$

where, in general, $\beta_j(w)$ is defined as in Remark 2.1. If $X_i = (1(T_i \geq t_1), \dots, 1(T_i \geq t_d))'$, as in Section 2, then $\Delta_{ij} > 0 \iff \Delta_{ij} = 1$. In this case, $\beta_j(w) = E[\beta_{ij}|W_i = w, \Delta_{ij} = 1]$, as defined in Section 2.

Taking limits as $e \rightarrow 0$, by the Dominated Convergence Theorem we obtain the key identifying equation

$$\delta_Y(w) = \beta(w)'\delta_X(w), \quad \text{for each } w \in \mathcal{W}.$$

²⁴Proof: this follows from the definition of the weights $\omega(W) = (E[\Lambda(W)]^{-1}\Lambda(W))$, which satisfy $E[\omega(W)] = I$.

The identified set for $\beta(w)$ is

$$\{\beta_0(w) + n(w) : \text{where } \delta_Y(w) = \beta_0(w)' \delta_X(w) \text{ and } n(w)' \delta_X(w) = 0\}, \quad (13)$$

where $\beta_0(w)$ is the true value that generated the data. Since there are an infinite number of n 's satisfying the conditions in (13), the result follows.

Proof of Proposition B.2: If there exists $w \in \mathcal{W}$ and j such that $\delta_{X_{-j}}(w) = 0$ and $\delta_{X_j}(w) \neq 0$. Then, $\delta_Y(w) = \beta_j(w) \delta_{X_j}(w)$ and hence $\beta_j(w)$ is identified as $\delta_Y(w) / \delta_{X_j}(w)$. The necessity follows from the next proposition applied to an ω equals to a Delta Dirac measure.

Proof of Proposition B.3: Without loss of generality take $j = 1$. From (13) a necessary and sufficient condition for identification of $E[\omega_1(W) \beta_1(W)]$ is $E[\omega_1(W) n_1(W)] = 0$ for all $n'(w) \delta_X(w) = 0$, where $n_1(w)$ denotes the first component of $n(w)$. Note that a basis for $\mathbb{R}^d$ can be constructed with first vector $\delta_X(w)$ and where the first elements for the rest of the vectors in the basis are $\delta_{X_j}(w)$ for $j = 2, \dots, d$. For example, $v_2(w) = (\delta_{X_2}(w), -\delta_{X_1}(w), 0, \dots, 0)'$ satisfies the orthogonality condition, and similarly we construct $v_3(w), \dots, v_d(w) = (\delta_{X_d}(w), 0, \dots, 0, -\delta_{X_1}(w))'$. Any linear combination of these vectors satisfy the orthogonality condition. Take $n(w) = \alpha(w) v_2(w)$, for a scalar $\alpha(w)$. Then, $n'(w) \delta_X(w) = 0$, and for identification we need $E[\omega_1(W) \alpha(W) \delta_{X_2}(W)] = 0$. Taking $\alpha(w) = \omega_1(w) \delta_{X_2}(w)$ conclude $\omega_1(w) \delta_{X_2}(w) = 0$ a.s. The same argument applied to v_j yields $\omega_1(w) \delta_{X_j}(w) = 0$ a.s. for $j = 3, \dots, d$. For the sufficiency note that by the previous arguments the first element $n_1(w)$ of a vector satisfying $n'(w) \delta_X(w) = 0$ can be written as linear combination of the $\delta_{X_j}(w)$, for $j = 2, \dots, d$, and thus $E[\omega_1(W) n_1(W)] = 0$.

Proof of Proposition B.4: From (13) a necessary and sufficient condition for identification of the TWLATE $E[\omega(W)' \beta(W)]$ is $E[\omega(W)' n(W)] = 0$ for all $n(w)' \delta_X(w) = 0$. The sufficiency is then clear, since $E[a(W) \delta_X(W)' n(W)] = 0$. For the necessity, use that any $n(W)$ satisfying $n(w)' \delta_X(w) = 0$ can be written as a linear combination of the vectors $v_j(w)$ for $j = 2, \dots, d$, say $n(w) = n_2(w) v_2(w) + \dots + n_d(w) v_d(w)$. Then, $0 = E[\omega(W)' n(W)] = E[\omega_2(W)' n_2(W)] + \dots + E[\omega_d(W)' n_d(W)]$, where $\omega_j(w) = \omega(w)' v_j(w)$. Taking $n_j(w) = \omega_j(w)$ we conclude $\omega_j(w) = 0$ a.s. for $j = 2, \dots, d$ and hence $\omega(w)$ must be proportional to $\delta_X(w)$, which shows the necessity.

D Asymptotic Theory

In this section we establish the asymptotic normality of the proposed estimator $\hat{\beta}$. Write $W_i = (W_{1i}, \dots, W_{mi})'$. Without loss of generality assume hereinafter that $z_0 = 0$. We introduce some

further notation and assumptions. Let $\varepsilon_{V_i} = V_i - E[V_i|Z_i, W_i]$ denote the regression errors for $V = Y$ and $V = X$. Define $\zeta_i = \varepsilon_{Y_i} - \bar{\beta}'_W \varepsilon_{X_i}$. Assume, for $V_i = Y_i$ and $V_i = X_{ij}$, $j = 1, \dots, d$,²⁵

$$E[V_i|Z_i, W_i] = \alpha_{0V}(Z_i) + \alpha_{1V}(Z_i)W_{i1} + \dots + \alpha_{mV}(Z_i)W_{im}. \quad (14)$$

For notational convenience, in the proofs we use the notation $H_i = Y_i(0)$ for the random intercept, rather than $\alpha_i = Y_i(0)$ (since α is used several times for generic coefficients). In the proofs we use the generic notation, for a generic V_i , $j, g = 1, \dots, m$,

$$\begin{aligned} \mu_{jg}^V(z) &= E[W_{ij}W_{ig}V_i|Z_i = z] \\ \sigma_{jg}^{2,V}(z) &= E[W_{ij}^2W_{ig}^2V_i^2|Z_i = z] \\ \kappa_{jg}^V(z) &= E[W_{ij}^3W_{ig}^3V_i^3|Z_i = z]. \end{aligned}$$

When V is identically 1, we drop the reference to V above.

We investigate the asymptotic properties of $\hat{\beta}$ under the following assumptions, which parallel those of [Hahn et al. \(2001\)](#):

Assumption 4 *Suppose that*

1. *The sample $\{\chi_i\}_{i=1}^n$ is an iid sample, where $\chi_i = (Y_i, X_i', W_i', Z_i)'$.*
2. *The density of Z , $f(\cdot)$, is continuous and bounded near $z = 0$. It is also bounded away from zero near $z = 0$. The matrices Γ and Δ_X , defined below, are positive definite and of full column rank, respectively.*
3. *The kernel k is continuous, symmetric and nonnegative-valued with compact support.*
4. *The functions $\mu_{jg}(z)$, $\mu_{jg}^X(z)$, $\mu_{jg}^Y(z)$, $\sigma_{jg}^2(z)$, $\sigma_{jg}^{2,X}(z)$, $\sigma_{jg}^{2,Y}(z)$, $\sigma_{jg}^{2,\zeta}(z)$, and $\kappa_{jg}^\zeta(z)$ are uniformly bounded near $z = 0$, with well-defined and finite left and right limits to $z = 0$, for $j, g = 1, \dots, m$.*

²⁵Here, we assume that $E[Y|Z, W]$ is a semiparametric function of W . If the support of W is discrete, say $\mathcal{W} = \{w_1, \dots, w_q\}$, as in our application, and we use in the estimation the vector $W = (1(W = w_1), \dots, 1(W = w_q))'$, this is a fully nonparametric, nonseparable model on W and Z .

Otherwise, if there is a desire for a fully nonparametric estimator in the continuous case, it can be done with minimal modifications to the estimator. Assume that $E[Y|Z, W] \in L_2$, and simply define the vector W used in the estimation as $(\varphi_1(W), \dots, \varphi_L(W))'$ where the φ_l are elements of a sieve basis in the space of squared integrable functions of W . The rest of the estimation procedure remains unchanged. The asymptotic results would need to be adapted to account for the fact that $L \rightarrow \infty$ as the sample increases, see e.g. [Chen \(2007\)](#). It is expected that this estimator converges at the same rate as the semiparametric estimator, because the additional nonparametric estimation is compensated by the averaging in W . Nevertheless, a formal proof of this is beyond the scope of this paper.

5. The bandwidth satisfies $nh_n^5 \rightarrow 0$.

6. For $V_i = Y_i$ and $V_i = X_{il}$ $l = 1, \dots, d$: (i) let equation (14) hold; for $j = 0, \dots, m$: (ii) for $z > 0$ or $z < 0$, $\alpha_{jV}(z)$ is twice continuously differentiable; (iii) there exists some $M > 0$ such that $\dot{\alpha}_{jV}^+(Z) = \lim_{u \downarrow Z} \partial \alpha_{jV}(u) / \partial u$ and $\ddot{\alpha}_{jV}^+(Z) = \lim_{u \downarrow Z} \partial^2 \alpha_{jV}(u) / \partial u^2$ are uniformly bounded on $(0, M]$. Similarly, $\dot{\alpha}_{jV}^-(Z) = \lim_{u \uparrow Z} \partial \alpha_{jV}(u) / \partial u$ and $\ddot{\alpha}_{jV}^-(Z) = \lim_{u \uparrow Z} \partial^2 \alpha_{jV}(u) / \partial u^2$ are uniformly bounded on $[-M, 0)$.

For a measurable function of the data $\varphi(\chi_i)$, define the local linear sample mean

$$\hat{E}[\varphi(\chi_i)] = \frac{1}{nh_n} \sum_{i=1}^n \varphi(\chi_i) k_{ih_n}.$$

Define $S_{+i} = D_i \cdot (1, Z_i, W_i', W_i' Z_i)'$, $S_{-i} = (1 - D_i) \cdot (1, Z_i, W_i', W_i' Z_i)'$, and $S_i = (S_{+i}', S_{-i}')'$. Recall $C = (1, W', Z, Z \cdot D, Z \cdot W', Z \cdot D \cdot W')'$. Note the spanning of the exogenous variables $(D_i, W_i' D_i, C_i)$ is the same as that of S_i , and thus the first stage is the regression of X_i on S_i . Let S_{in} and C_{in} be defined the same as S_i and C_i but with Z_i replaced by Z_i/h_n . Define $\tilde{X}_i = (X_i', C_i')'$, $\tilde{X}_{in} = (X_i', C_{in}')'$, and $\tau = (\bar{\beta}_W', \eta')'$, where $\eta = (\eta_1, \eta_2', \eta_3, \eta_4, \eta_5', \eta_6')'$ has the same dimension as C_i and is such that

$$\eta' C_i = \eta_1 + \eta_2' W_i + \eta_3 Z_i + \eta_4 D_i Z_i + \eta_5' W_i Z_i + \eta_6' W_i D_i Z_i.$$

Define $\tau_n = (\bar{\beta}_W', \eta_1, \eta_2', \eta_3 h_n, h_n \eta_4, h_n \eta_5', h_n \eta_6')'$, so that $\tilde{X}_i' \tau = \tilde{X}_{in}' \tau_n$. With this notation in place, the 2SLS is the first $d \times 1$ subcomponent of

$$\begin{aligned} \hat{\tau}_n &= \left(\hat{E} [\tilde{X}_{in} S_{in}'] \left(\hat{E} [S_{in} S_{in}'] \right)^{-1} \hat{E} [S_{in} \tilde{X}_{in}] \right)^{-1} \hat{E} [\tilde{X}_{in} S_{in}'] \left(\hat{E} [S_{in} S_{in}'] \right)^{-1} \hat{E} [S_{in} Y_i] \\ &= \tau_n + \left(\hat{E} [\tilde{X}_{in} S_{in}'] \left(\hat{E} [S_{in} S_{in}'] \right)^{-1} \hat{E} [S_{in} \tilde{X}_{in}] \right)^{-1} \hat{E} [\tilde{X}_{in} S_{in}'] \left(\hat{E} [S_{in} S_{in}'] \right)^{-1} \hat{E} [S_{in} U_i], \end{aligned}$$

where $U_i = Y_i - \tilde{X}_{in}' \tau_n = Y_i - \tilde{X}_i' \tau$.

We show in Lemma D.11 that

$$\sqrt{nh_n} (\hat{\tau}_n - \tau_n) \rightarrow_d N(0, \Omega), \quad (15)$$

and provide an expression for Ω . The asymptotic normality for $\sqrt{nh_n} (\hat{\beta}_n - \bar{\beta}_W)$ then follows as

$$\sqrt{nh_n} (\hat{\beta}_n - \bar{\beta}_W) \rightarrow_d N(0, \Sigma),$$

where $\Sigma = e_0' \Omega e_0$, and e_0 is a selection matrix that selects the first d elements of $\hat{\tau}_n$.

We introduce some notation that will be used throughout,

$$\gamma_l = \int_0^\infty u^l k(u) du,$$

$$\mu_{jg}^{+,V} = \lim_{z \downarrow 0} E[W_{ij} W_{ig} V_i | Z_i = z] \quad \mu_{j\rho}^{-,V} = \lim_{z \uparrow 0} E[W_{ij} W_{ig} V_i | Z_i = z],$$

$$k_{ih_n}^+ = k(Z_i/h_n)1(Z_i \geq 0) \quad k_{ih_n}^- = k(Z_i/h_n)1(Z_i < 0),$$

$$b_{ljg}^+ = \gamma_l \mu_{jg}^+, \quad b_{ljg}^- = \gamma_l \mu_{jg}^-, \quad b_{ljg}^{+,X} = \gamma_l \mu_{jg}^{+,X}, \quad b_{ljg}^{-,X} = \gamma_l \mu_{jg}^{-,X}.$$

We also use the notation that dropping an index means dropping the variable associated to the index, so for example, $\mu_j^{+,V} = \lim_{z \downarrow 0} E[W_{ij} V_i | Z_i = z]$ and $\mu^{+,V} = \lim_{z \downarrow 0} E[V_i | Z_i = z]$, and when V is dropped from the notation it means that it is substituted by 1. The notation $\mu_W^{+,V} = (\mu_1^{+,V}, \dots, \mu_m^{+,V})$ means a row vector with the same dimension as W , and $\mu_\otimes^{+,V}$ is the $m \times m$ matrix with elements $\mu_{jg}^{+,V}$. Similarly, we denote $b_{lW}^{+,X} = \gamma_l \mu_W^{+,X}$ and $b_{l\otimes}^{+,X} = \gamma_l \mu_\otimes^{+,X}$, and likewise for other quantities.

Lemma D.1 *Under Assumption 4 1-5,*

$$\hat{E} \left[\tilde{X}_{in} S'_{in} \right] \rightarrow_p \Delta_X,$$

where

$$\Delta_X = f(0) \begin{bmatrix} b_{00}^{+,X} & b_{10}^{+,X} & b_{0W}^{+,X} & b_{1W}^{+,X} & b_{00}^{-,X} & b_{10}^{-,X} & b_{0W}^{-,X} & b_{1W}^{-,X} \\ b_{00}^+ & b_{10}^+ & b_{0W}^+ & b_{1W}^+ & b_{00}^- & b_{10}^- & b_{0W}^- & b_{1W}^- \\ b_{0W}^{+'} & b_{1W}^{+'} & b_{0\otimes}^+ & b_{1\otimes}^+ & b_{0W}^{-'} & b_{1W}^{-'} & b_{0\otimes}^- & b_{1\otimes}^- \\ b_1^+ & b_2^+ & b_{1W}^+ & b_{2W}^+ & b_1^- & b_2^- & b_{1W}^- & b_{2W}^- \\ b_1^+ & b_2^+ & b_{1W}^+ & b_{2W}^+ & 0 & 0 & 0 & 0 \\ b_{1W}^{+'} & b_{2W}^{+'} & b_{1\otimes}^+ & b_{2\otimes}^+ & b_{1W}^{-'} & b_{2W}^{-'} & b_{1\otimes}^- & b_{2\otimes}^- \\ b_{1W}^{+'} & b_{2W}^{+'} & b_{1\otimes}^+ & b_{2\otimes}^+ & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Proof. Let

$$\tau_{ljg}^+ = \frac{1}{nh_n} \sum_{i=1}^n \left(\frac{Z_i}{h_n} \right)^l W_{ij} W_{ig} k_{ih_n}^+, \quad l, \rho = 0, 1, 2, j = 1, \dots, m.$$

Then, by the change of variables $u = Z/h_n$,

$$\begin{aligned} E[\tau_{l j g}^+] &= h_n^{-1} E \left[\left(\frac{Z_i}{h_n} \right)^l W_{ij} W_{ig} k_{ih_n}^+ \right] \\ &= \int_0^\infty u^l k(u) \mu_{jg}(uh_n) f(uh_n) du \\ &= f(0) \gamma_l \mu_{jg}^+ + o(1), \end{aligned}$$

where $\mu_{jg}(z) = E[W_{ij} W_{ig} | Z_i = z]$ and the convergence follows by the Dominated Convergence theorem. As for the variance

$$\begin{aligned} \text{Var}(\tau_{l j g}^+) &\leq (nh_n^2)^{-1} E \left[\left(\frac{Z_i}{h_n} \right)^{2l} W_{ij}^2 W_{ig}^2 k_{ih_n}^{+2} \right] \\ &= (nh_n)^{-1} \int_0^\infty u^{2l} k^2(u) \sigma_{jg}^2(uh_n) f(uh_n) du \\ &= o(1), \end{aligned}$$

again by the Dominated Convergence theorem.

Similarly, define

$$\tau_{l j g}^{+,X} = \frac{1}{nh_n} \sum_{i=1}^n \left(\frac{Z_i}{h_n} \right)^l W_{ij} W_{ig} X_i k_{ih_n}^+, \quad l = 0, 1, 2, j, g = 1, \dots, m.$$

Then, by the change of variables $u = Z/h_n$,

$$\begin{aligned} E[\tau_{l j g}^{+,X}] &= h_n^{-1} E \left[\left(\frac{Z_i}{h_n} \right)^l W_{ij} W_{ig} X_i k_{ih_n}^+ \right] \\ &= \int_0^\infty u^l k(u) \mu_{jg}^X(uh_n) f(uh_n) du \\ &= f(0) \gamma_l \mu_{jg}^{+,X} + o(1), \end{aligned}$$

where $\mu_{jg}^X(z) = E[W_{ij} W_{ig} X_i | Z_i = z]$, by the Dominated Convergence theorem. As for the vari-

ance, when $d = 1$,

$$\begin{aligned} \text{Var}(\tau_{ljg}^{+,X}) &\leq (nh_n^2)^{-1} E \left[\left(\frac{Z_i}{h_n} \right)^{2l} W_{ij}^2 W_{ig}^2 X_i^2 k_{ih_n}^{+2} \right] \\ &= (nh_n)^{-1} \int_0^\infty u^{2l} k^2(u) \sigma_{jg}^{2,X}(uh_n) f(uh_n) du \\ &= o(1), \end{aligned}$$

again by the Dominated Convergence theorem. The same argument holds when $d > 1$. The proof for τ_{ljg}^- and $\tau_{ljg}^{-,X}$, which replace $k_{ih_n}^+$ by $k_{ih_n}^-$, is analogous, and hence omitted. ■

Lemma D.2 Under Assumption 4 1-5,

$$\hat{E}[S_{in} S'_{in}] \rightarrow_p \Gamma$$

where

$$\Gamma = f(0) \begin{bmatrix} b_0^+ & b_1^+ & b_{0W}^+ & b_{1W}^+ & 0 & 0 & 0 & 0 \\ b_1^+ & b_2^+ & b_{1W}^+ & b_{2W}^+ & 0 & 0 & 0 & 0 \\ b_{0W}^{+'} & b_{1W}^{+'} & b_{0\otimes}^+ & b_{1\otimes}^+ & 0 & 0 & 0 & 0 \\ b_{1W}^{+'} & b_{2W}^{+'} & b_{1\otimes}^+ & b_{2\otimes}^+ & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & b_0^- & b_1^- & b_{0W}^- & b_{1W}^- \\ 0 & 0 & 0 & 0 & b_1^- & b_2^- & b_{1W}^- & b_{2W}^- \\ 0 & 0 & 0 & 0 & b_{0W}^{-'} & b_{1W}^{-'} & b_{0\otimes}^- & b_{1\otimes}^- \\ 0 & 0 & 0 & 0 & b_{1W}^{-'} & b_{2W}^{-'} & b_{1\otimes}^- & b_{2\otimes}^- \end{bmatrix}.$$

Proof. The proof is analogous to that of Lemma D.1 and hence is omitted. ■

Lemma D.3 Under Assumption 4 1-5,

$$\hat{E}[S_{in} Y_i] \rightarrow_p \Delta_Y,$$

where $\Delta_Y = f(0)(b_{00}^{+,Y}, b_{10}^{+,Y}, b_{0W}^{+,Y}, b_{1W}^{+,Y}, b_{00}^{-,Y}, b_{10}^{-,Y}, b_{0W}^{-,Y}, b_{1W}^{-,Y})'$.

Proof. The proof is analogous to that of Lemma D.1 and hence is omitted. ■

Lemma D.4 Assume 3, 2 and 4. Then,

$$\Delta_Y = \Delta_X' \tau_0,$$

where $\tau_0 = (\bar{\beta}_W', \eta_1, \eta_2', 0, 0, 0', 0')'$.

Proof. By Assumption 4 6, we can write

$$E[H_i|W_i, Z_i] = \alpha_{0H}(Z_i) + \alpha_{1H}(Z_i)W_{1i} + \cdots + \alpha_{mH}(Z_i)W_{mi}.$$

Take for example $b_{00}^{+,Y}$. By substitution of the random coefficients representation and the identifying assumptions

$$b_{00}^{+,Y} = \bar{\beta}_W b_{00}^{+,X} + \eta_1 b_{00}^+ + \eta_2' b_{0W}^+,$$

where

$$\eta_1 = \alpha_{0H}(0)$$

and

$$\eta_2 = (\alpha_{1H}(0), \dots, \alpha_{mH}(0))'.$$

The same argument applied to other components of Δ_Y , lead to the desired equality $\Delta_Y = \Delta_X' \tau_0$. ■

The previous Lemmas show the consistency of $\hat{\beta}$, since

$$\hat{\tau}_n \rightarrow_p (\Delta_X \Gamma^{-1} \Delta_X')^{-1} \Delta_X \Gamma^{-1} \Delta_Y = \tau_0$$

We prove several Lemmas that will yield the asymptotic normality of $\sqrt{nh_n}(\hat{\tau}_n - \tau_n)$, and hence that of $\sqrt{nh_n}(\hat{\beta}_n - \bar{\beta}_W)$.

Henceforth, to simplify notation we consider $m = 1$, although we indicate the necessary changes in the arguments for $m > 1$. Define the function

$$\begin{aligned} \zeta_H(Z, W) &= \alpha_{0H}(Z) + \alpha_{1H}(Z)W - \alpha_{0H}^+ + \alpha_{1H}^+ W \\ &\quad - (\dot{\alpha}_{0H}^+ + \dot{\alpha}_{1H}^+ W) Z - \frac{1}{2} (\ddot{\alpha}_{0H}^+ + \ddot{\alpha}_{1H}^+ W) Z^2, \end{aligned}$$

where $\dot{\alpha}_{0H}^+ = \lim_{z \downarrow 0} \partial \alpha_{0H}(z) / \partial z$ and $\ddot{\alpha}_{0H}^+ = \lim_{z \downarrow 0} \partial^2 \alpha_{0H}(z) / \partial z^2$, and similarly for α_{1H} . We use later that

$$\sup_{0 < z < Mh_n} |\zeta_H(z, w)| = o(h_n^2)(1 + |w|).$$

The function $\zeta_H(z, w)$ is the Taylor's remainder of order two of $E[H_i|W_i = w, Z = z] := \alpha_{0H}(z) + \alpha_{1H}(z)w$ around $z = 0$. We can also relate the coefficients in this expansion with the coefficients in η . Following the arguments above, it can be shown that

$$\tilde{\eta} = \arg \min_{\eta} \sum_{i=1}^n (H_i - \eta' C_i)^2 k_{h_n}(Z_i).$$

estimates consistently η . Thus,

$$\begin{aligned}\eta_1 &= \alpha_{0H}(0), \quad \eta_2 = \alpha_{1H}(0), \quad \eta_3 = \dot{\alpha}_{0H}(0), \\ \eta_4 &= 0, \quad \eta_5 = \dot{\alpha}_{1H}(0), \quad \eta_6 = 0.\end{aligned}$$

Recall $U_i = Y_i - \tilde{X}_i' \tau$. Then, from the definitions above

$$\begin{aligned}E[U_i|W_i, Z_i] &= E[H_i|W_i, Z_i] - \alpha_{0H}^+ + \alpha_{1H}^+ W - \dot{\alpha}_{0H}^+ Z + \dot{\alpha}_{1H}^+ W Z \\ &= \frac{1}{2} (\ddot{\alpha}_{0H}^+ + \ddot{\alpha}_{1H}^+ W_i) Z_i^2 + \zeta_H(Z_i, W_i).\end{aligned}$$

The following Lemmas make use of Assumptions 3, 2 and 4. Define $\mu_\rho(z) = E[W_{ij}^\rho | Z_i = z]$, for $\rho = 1, 2$, and their one-sided versions as usual.

Lemma D.5 (Numerator: Expectation)

$$E \left[\frac{1}{nh_n} \sum_{i=1}^n S_{in} U_i k_{ih_n} \right] \rightarrow_p \frac{1}{2} f(0) h_n^2 (b_U + o(1)),$$

where

$$b_U = \begin{bmatrix} \gamma_2(\ddot{\alpha}_{0H}^+ + \ddot{\alpha}_{0H}^+ \mu_1^+) \\ \gamma_3(\ddot{\alpha}_{0H}^+ + \ddot{\alpha}_{0H}^+ \mu_1^+) \\ \gamma_2(\ddot{\alpha}_{0H}^+ \mu_1^+ + \ddot{\alpha}_{0H}^+ \mu_2^+) \\ \gamma_3(\ddot{\alpha}_{0H}^+ \mu_1^+ + \ddot{\alpha}_{0H}^+ \mu_2^+) \\ \gamma_2(\ddot{\alpha}_{0H}^- + \ddot{\alpha}_{0H}^+ \mu_1^-) \\ \gamma_3(\ddot{\alpha}_{0H}^- + \ddot{\alpha}_{0H}^+ \mu_1^-) \\ \gamma_2(\ddot{\alpha}_{0H}^- \mu_1^- + \ddot{\alpha}_{0H}^- \mu_2^+) \\ \gamma_3(\ddot{\alpha}_{0H}^- \mu_1^- + \ddot{\alpha}_{0H}^- \mu_2^+) \end{bmatrix}.$$

Proof. Let

$$u_{l\rho}^+ = \frac{1}{nh_n} \sum_{i=1}^n \left(\frac{Z_i}{h_n} \right)^l W_i^\rho U_i k_{ih_n}^+, \quad l, \rho = 0, 1.$$

Then, write

$$\begin{aligned}
E[u_{l\rho}^+] &= h_n^{-1} E \left[\left(\frac{Z_i}{h_n} \right)^l W_i^\rho U_i k_{ih_n}^+ \right] \\
&= h_n^{-1} E \left[\left(\frac{Z_i}{h_n} \right)^l W_i^\rho \left(\frac{1}{2} (\ddot{\alpha}_{0H}^+ + \ddot{\alpha}_{1H}^+ W_i) Z_i^2 + \zeta_H(Z_i, W_i) \right) k_{ih_n}^+ \right] \\
&= h_n^{-1} \frac{1}{2} \ddot{\alpha}_{0H}^+ E \left[\left(\frac{Z_i}{h_n} \right)^l W_i^\rho Z_i^2 k_{ih_n}^+ \right] + h_n^{-1} \frac{1}{2} \ddot{\alpha}_{1H}^+ E \left[\left(\frac{Z_i}{h_n} \right)^l W_i^{\rho+1} Z_i^2 k_{ih_n}^+ \right] \\
&\quad + h_n^{-1} E \left[\left(\frac{Z_i}{h_n} \right)^l W_i^\rho \zeta_H(Z_i, W_i) k_{ih_n}^+ \right].
\end{aligned}$$

By the change of variables $u = Z/h_n$,

$$\begin{aligned}
h_n^{-1} E \left[\left(\frac{Z_i}{h_n} \right)^l W_i^\rho Z_i^2 k_{ih_n}^+ \right] &= h_n^2 \int_0^\infty u^{l+2} k(u) \mu_\rho(uh_n) f(uh_n) du \\
&= h_n^2 \mu_\rho^+ f(0^+) \gamma_{l+2} + o(1),
\end{aligned}$$

and similarly

$$h_n^{-1} E \left[\left(\frac{Z_i}{h_n} \right)^l W_i^{\rho+1} Z_i^2 k_{ih_n}^+ \right] = h_n^2 \mu_{\rho+1}^+ f(0^+) \gamma_{l+2} + o(1).$$

On the other hand, assume without loss of generality that $[-M, M]$ contains the support of k , so that

$$h_n^{-1} E \left[\left(\frac{Z_i}{h_n} \right)^l W_i^\rho \zeta_H(Z_i, W_i) k_{ih_n}^+ \right] = o(h_n^2).$$

The proof for the left limit version is analogous, and hence omitted. ■

Lemma D.6 (Numerator: Conditional Expectation)

$$\frac{1}{nh_n} \sum_{i=1}^n E[S_{in} U_i k_{ih_n} | Z_i, W_i] = \frac{1}{nh_n} \sum_{i=1}^n E[S_{in} U_i k_{ih_n}] + o_p(h_n^2).$$

Proof. We have

$$\begin{aligned}
\frac{1}{nh_n} \sum_{i=1}^n E[S_{in} U_i k_{ih_n} | Z_i, W_i] &= \frac{1}{nh_n} \sum_{i=1}^n S_{in} k_{ih_n} \left(\frac{1}{2} (\ddot{\alpha}_{0H}^+ + \ddot{\alpha}_{1H}^+ W_i) Z_i^2 + \zeta_H(Z_i, W_i) \right) \\
&= \frac{1}{2} \ddot{\alpha}_{0H}^+ \frac{1}{nh_n} \sum_{i=1}^n S_{in} k_{ih_n} Z_i^2 + \frac{1}{2} \ddot{\alpha}_{1H}^+ \frac{1}{nh_n} \sum_{i=1}^n S_{in} k_{ih_n} W_i Z_i^2 \\
&\quad + \frac{1}{nh_n} \sum_{i=1}^n S_{in} k_{ih_n} \zeta_H(Z_i, W_i).
\end{aligned}$$

Observe that

$$\begin{aligned}
Var \left(\frac{1}{nh_n} \sum_{i=1}^n S_{in} k_{ih_n} Z_i^2 \right) &= (nh_n^2)^{-1} Var (S_{in} k_{ih_n} Z_i^2) \\
&\leq C (nh_n)^{-1} h_n^{-1} E [S_{in} S_{in}' k_{ih_n}^2 Z_i^4] \\
&= O((nh_n)^{-1} h_n^4) \\
&= o(1),
\end{aligned}$$

since for $l, \rho = 0, 1, 2$

$$\begin{aligned}
h_n^{-1} E \left[\left(\frac{Z_i}{h_n} \right)^l W_i^\rho k_{ih_n}^{+2} Z_i^4 \right] &= h_n^4 \int_0^\infty u^l k^2(u) \mu_\rho(u h_n) f(u h_n) du \\
&= h_n^4 \mu_\rho^+ f(0^+) v_l + o(1),
\end{aligned}$$

where

$$v_l = \int_0^\infty u^l k^2(u) du,$$

and similarly

$$h_n^{-1} E \left[\left(\frac{Z_i}{h_n} \right)^l W_i^\rho k_{ih_n}^{-2} Z_i^4 \right] = h_n^4 \mu_\rho^- f(0^-) v_l + o(1).$$

Likewise,

$$Var \left(\frac{1}{nh_n} \sum_{i=1}^n S_{in} k_{ih_n} W_i Z_i^2 \right) = o(1).$$

and

$$Var \left(\frac{1}{nh_n} \sum_{i=1}^n S_{in} k_{ih_n} \zeta_H(Z_i, W_i) \right) = o(1).$$

Note that

$$\frac{1}{nh_n} \sum_{i=1}^n S_{in} k_{ih_n}^+ (U_i - E[U_i|Z_i, W_i]) = \frac{1}{nh_n} \sum_{i=1}^n S_{in} k_{ih_n}^+ \zeta_i,$$

where $\zeta_i = U_i - E[U_i|Z_i, W_i] = \varepsilon_{Y_i} - \bar{\beta}_W \varepsilon_{X_i}$ denotes the regression error. Then, we have the following result. ■

Lemma D.7 (Numerator: Conditional Variance)

$$\text{Var} \left(\frac{1}{nh_n} \sum_{i=1}^n S_{+in} k_{ih_n} \zeta_i \right) = \frac{1}{nh_n} \Sigma_{U^+} + o(1),$$

where

$$\Sigma_{U^+} = f(0^+) \begin{bmatrix} v_0 & v_2 & v_0 \sigma_1^{2+} & v_2 \sigma_1^{2+} \\ v_2 & v_4 & v_2 \sigma_1^{2+} & v_2 \sigma_2^{2+} \\ v_0 \sigma_1^{2+'} & v_2 \sigma_1^{2+'} & v_0 \text{diag}(\sigma_2^{2+}) & v_4 \sigma_1^{2+'} \\ v_2 \sigma_1^{2+'} & v_2 \sigma_2^{2+'} & v_4 \sigma_1^{2+'} & v_4 \text{diag}(\sigma_2^{2+}) \end{bmatrix},$$

$$v_l = \int_0^\infty u^l k^2(u) du, \quad \sigma_{j\rho}^{2+} = \lim_{z \downarrow 0} E[W_{ji}^{2\rho} \zeta_i^2 | Z = z] \text{ and } \sigma_\rho^{2+} = (\sigma_{1\rho}^{2+}, \dots, \sigma_{c\rho}^{2+}).$$

Proof. Consider the generic term, for $l, \rho = 0, 1, j = 1, \dots, c$,

$$\frac{1}{nh_n} \sum_{i=1}^n \left(\frac{Z_i}{h_n} \right)^l W_{ji}^\rho k_{ih_n}^+ \zeta_i,$$

and its variance, which equals

$$\begin{aligned} (nh_n)^{-1} h_n^{-1} E \left[\left(\frac{Z_i}{h_n} \right)^{2l} W_{ji}^{2\rho} k_{ih_n}^{+2} \zeta_i^2 \right] &= (nh_n)^{-1} \int_0^\infty u^{2l} k^2(u) \sigma_{j\rho}^2(uh_n) f(uh_n) du \\ &= (nh_n)^{-1} f(0^+) \sigma_{j\rho}^{2+} v_{2l} + o(1) \end{aligned}$$

where $\sigma_{j\rho}^2(z) = E[W_{ji}^{2\rho} \zeta_i^2 | Z_i = z]$. Similarly, we have the following result, which proof is the same as in the previous lemma. ■

Lemma D.8

$$\text{Var} \left(\frac{1}{nh_n} \sum_{i=1}^n S_{-in} k_{ih_n} \zeta_i \right) = \frac{1}{nh_n} \Sigma_{U^-} + o(1),$$

where

$$\Sigma_{U^-} = f(0^-) \begin{bmatrix} v_0 & v_2 & v_0\sigma_1^{2-} & v_2\sigma_1^{2-} \\ v_2 & v_4 & v_2\sigma_1^{2-} & v_2\sigma_2^{2-} \\ v_0\sigma_1^{2-'} & v_2\sigma_1^{2-'} & v_0\text{diag}(\sigma_2^{2-}) & v_4\sigma_1^{2-'} \\ v_2\sigma_1^{2-'} & v_2\sigma_2^{2-'} & v_4\sigma_1^{2-'} & v_4\text{diag}(\sigma_2^{2-}) \end{bmatrix},$$

$$\sigma_{j\rho}^{2-} = \lim_{z \downarrow 0} E[W_{ji}^{2\rho} \zeta_i^2 | Z = z] \text{ and } \sigma_\rho^{2-} = (\sigma_{1\rho}^{2-}, \dots, \sigma_{c\rho}^{2-}).$$

Define

$$\Sigma_U = \begin{bmatrix} \Sigma_{U^+} & 0 \\ 0 & \Sigma_{U^-} \end{bmatrix}$$

Lemma D.9 (Numerator: Conditional CLT)

$$(nh_n)^{-1/2} \sum_{i=1}^n S_{in} k_{ih_n} \zeta_i \rightarrow_d N(0, \Sigma_U).$$

Proof. Consider a generic term for $l, \rho = 0, 1, j = 1, \dots, c$,

$$\frac{1}{\sqrt{nh_n}} \sum_{i=1}^n \left(\frac{Z_i}{h_n} \right)^l W_{ji}^\rho k_{ih_n}^+ \zeta_i.$$

We apply Lyapounov with third absolute moment. By the lemma on the asymptotic variance, we need to establish

$$(nh_n)^{-1/2} h_n^{-1} E \left[\left(\frac{Z_i}{h_n} \right)^{3l} W_{ji}^{3\rho} k_{ih_n}^{+3} \zeta_i^3 \right] = o(1).$$

But note that, defining $\kappa_{j\rho}(z) = E[W_{ji}^{3\rho} \zeta_i^3 | Z = z]$,

$$\begin{aligned} h_n^{-1} E \left[\left(\frac{Z_i}{h_n} \right)^{3l} W_{ji}^{3\rho} k_{ih_n}^{+3} \zeta_i^3 \right] &= \int_0^\infty u^{3l} k^3(u) \kappa_{j\rho}(uh_n) f(uh_n) du \\ &= O(1). \end{aligned}$$

The same holds for the left limit part. ■

Lemma D.10 (Numerator: Unconditional CLT)

$$(nh_n)^{-1/2} \sum_{i=1}^n S_{in} U_i k_{ih_n} - \frac{(nh_n)^{1/2} h_n^2}{2} f(0) b_U \rightarrow_d N(0, \Sigma_U).$$

Proof. It follows from previous Lemmas. ■

Lemma D.11 (Main CLT)

$$\sqrt{nh_n}(\hat{\tau}_n - \tau_n) \rightarrow_d N(0, \Omega),$$

where

$$\Omega = (\Delta\Gamma^{-1}\Delta')^{-1} \Delta\Gamma^{-1}\Sigma_U\Gamma^{-1}\Delta' (\Delta\Gamma^{-1}\Delta')^{-1}.$$

Proof. It follows from previous Lemmas. ■